

SPOTLIGHT: Title is TBD

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Abstract

1 Introduction

Hybrid spot/on-demand strategies [21, 57] have been extensively studied.

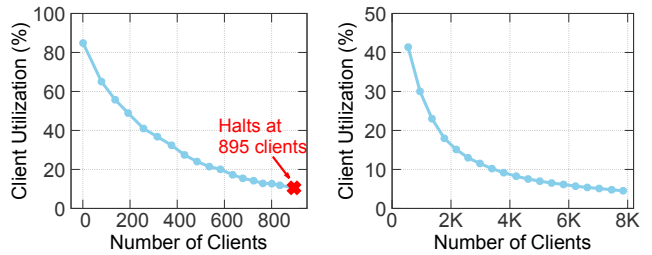
2 Motivation

FL trains models without centralizing data, and its deployments have grown accordingly. Production systems already serve tens of millions of devices [7]. Each device returns a model update that must be merged into the global model. The aggregation layer must handle all of this work. Existing FL frameworks, including Flower [6], IBM FL [33], FATE [32], and Papaya [17], implement this layer as a single aggregator that manages the global model. We examine two challenges: the scalability limits of a single aggregator (§2.1) and the infrastructure cost of scaling the aggregation tier (§2.2).

2.1 Scaling FL aggregation

In FL, an aggregator is responsible for three operations [7, 17]. It manages the lifecycle of connecting clients, selects and forwards devices into training rounds, and incorporates the returned updates into the global model [20]. Deployments now reach tens of millions of devices [7]. Building an aggregation layer for that scale places significant load on the aggregator, as the demand of all three operations rises with the number of clients.

Aggregators incorporate client updates either synchronously or asynchronously. Flower [6] adopts synchronous client updates, requiring a quorum of clients before each round. On the other hand, Papaya [17] uses asynchronous client updates, applying each update on arrival without waiting for a quorum. To see how each behaves as the client population increases, we measure client utilization. Fig. 1 shows the result with a 10MB payload. Flower’s utilization (Fig. 1a) decreases from 85% to below 12%, and at 895 clients



(a) Flower, 100 to 895 clients. (b) Papaya, 500 to 8,000 clients.

Figure 1: **Client utilization degrades as the client population grows in both Flower and Papaya FL systems.**

it halts entirely. Papaya sustains a much larger population, but its utilization (Fig. 1b) decreases from 41% to near zero by 8,000 clients. The two systems fail differently. Flower halts, while Papaya keeps running but no longer benefits from additional clients. We could confirm that both systems reach a threshold at which a single aggregator can no longer process the load.

An intuitive approach to address this challenge is to add more aggregators. In a stateless service, this approach can be sufficient, as any instance can serve any request. However, FL aggregation is *stateful*: each aggregator maintains a running model that reflects only the updates it has received. Therefore, partitioning clients across instances also partitions the model. Each instance produces a sub-global model that is accurate on its own subset of clients but incomplete for the rest. Adding or replicating the aggregator alone is not enough, as it can yield several partial models and no global one. Creating and maintaining a single global model requires an explicit step that merges the sub-global models.

The merging step can be performed within the aggregator layer, with the aggregators combining their sub-global models. But this approach requires every aggregator to coordinate with every other, and the coordination overhead increases with the number of aggregators. Placing the merge in a separate tier removes this overhead, as each tier then coordinates only

within itself. Each lower-tier aggregator handles a subset of clients, so its load fluctuates with client participation. The upper-tier aggregator merges the sub-global models, so its load depends on the number of lower-tier aggregators rather than on the client count.

Furthermore, because these loads change throughout FL operation, the number of aggregators in each tier cannot be fixed in advance. The system, therefore, should provision and deprovision aggregators as the load changes. When and how to perform these operations should be carefully designed, in addition to introducing the merging tier.

Design Insight #1: Scaling FL aggregation requires a two-tier hierarchy, where a lower tier aggregates client updates and an upper tier merges the resulting sub-global models. Because load varies throughout FL operation, this hierarchy must also be elastic, and deciding when and how to scale it is an additional design challenge.

2.2 The cost of FL running at scale

[the cost of scalable FL] Even after the scalability challenge is addressed, another challenge remains. In practice, the aggregation layer runs on always-on, on-demand cloud VMs [7, 42, 52], and scaling it to serve large client populations makes the cloud cost prohibitive. To quantify this cost, we measure the per-round aggregator time for training ResNet-18 on CIFAR-100 with FedAvg [36] on an AWS p3.2xlarge instance, and the median round takes 732 seconds. Prior work reports that this task converges in 1,198 rounds [49], so a full run requires roughly 244.5 hours of aggregator time. At the on-demand rate of \$3.06 per hour, this amounts to \$748.25 just to train a single model to convergence.

[spot is cheap] Spot instances offer a much cheaper alternative, typically 3 – 10 $\times$ cheaper than on-demand [18, 48, 56]. The caveat is that the provider can reclaim a spot instance at any time, causing the aggregator to lose its progress. To estimate the potential cost advantage, we first consider the ideal case in which no preemption occurs. In that case, the same 244.5-hour run at the spot rate of \$0.345 per hour [13, 27] will cost only \$84.36, an 88.7% reduction over on-demand.

[reality in spot] In practice, however, spot preemptions do occur. Without proper fault-tolerance mechanisms, a preemption reclaims the spot VM, causing the aggregator to lose all progress. To continue the FL process, a new instance must be provisioned and the computation restarted from scratch. We simulate spot preemptions on the same training task, using a real availability trace [56] and restarting on every preemption. Averaged over ten runs, the aggregator completes only 156 of the 1,198 rounds within the 244.5-hour ideal window, so finishing the full run would take roughly 1,875 hours on spot instances. At \$0.345 per hour, this comes to \$646.88, just 13.5% below the on-demand cost. Fig. 2 compares the three cases. Even if we disregard the effect of preemptions on

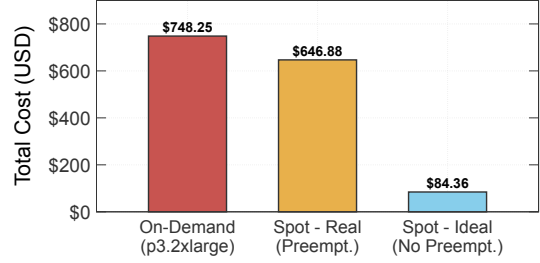


Figure 2: **Aggregator cost to train ResNet-18 on CIFAR-100 to convergence on a p3.2xlarge instance (1,198 rounds [49]).** Without fault tolerance, preemptions force full restarts, raising the spot cost from an ideal \$84.36 to \$646.88, close to the on-demand cost of \$748.25.

model accuracy, spot instances without fault tolerance offer only a negligible cost advantage over on-demand.

[need for FT] Recovering the cost advantage of spot instances therefore requires fault tolerance. Periodic checkpointing saves the model so that a preempted run can resume from the last checkpoint [xxx₁, (IKK₁: chkpnt papers)], but determining an optimal checkpoint interval is non-trivial and adds implementation complexity. Aggregator relocation moves the aggregator to another instance before the current spot VM is reclaimed [xxx₂], but provisioning a new instance takes time and delays aggregation. Providers send a termination signal shortly before reclaiming an instance (e.g., 2 minutes [3] or 30 seconds [38]), but the termination notice is short, only two minutes on AWS and thirty seconds on GCP. These techniques have not been applied to or optimized for FL aggregation on spot instances. Therefore, we address fault tolerance for FL on spot not only by adapting them to the setting, but also by developing new mechanisms designed for FL aggregation.

Design Insight #2: The cost advantage of FL aggregation on spot instances is achievable only with fault tolerance. This fault tolerance requires both applying well-established principles and developing new techniques tailored to FL aggregation.

3 SPOTLIGHT System Overview

SPOTLIGHT enables federated learning to scale elastically on spot infrastructure without sacrificing aggregator performance or cost efficiency. This section overviews the architecture of SPOTLIGHT (§3.1) and the three design principles behind it (§3.2): *two-tiered aggregation* that scopes each aggregator to its assigned clients and combines partial models into a consistent global model, *proactive fault tolerance* that migrates workloads and checkpoints aggregator state upon detecting failure likelihood, and *elastic spot provisioning* that treats scalability and fault tolerance as co-equal concerns to fully exploit spot infrastructure.

3.1 SPOTLIGHT Architecture

Fig. 3 shows the architecture of SPOTLIGHT. The system comprises three main components: the **Resource Manager (RM)**, the **aggregation layer**, and the **Performance Manager**. The aggregation layer consists of **Sub-Global Aggregators**, **Global Aggregators**, and **Failover Edge Aggregators**, which extend the aggregation hierarchy to the edge when cloud aggregators are unavailable. All external client connections are fronted by Envoy, and MongoDB persists coordination state shared across all components.

The RM is itself horizontally scalable, deployed as multiple Kubernetes instances, and internally comprises four subcomponents. The **Scheduler** manages client membership, assigns clients to Sub-Global Aggregator cohorts, and schedules incoming updates. The **Scaler** provisions and deprovisions Sub-Global and Global Aggregators in response to signals from the Performance Manager, forming a closed loop that drives elastic scaling. The **Migrator** monitors aggregator failure likelihood and, upon detecting an at-risk instance, determines an appropriate migration target, either a healthy cloud aggregator or a Failover Edge Aggregator, and triggers workload migration and state checkpointing on the at-risk instance. The **Leader Elector** runs Raft consensus across RM instances to elect a single leader with sole authority over scaling and migration decisions, allowing the RM to scale horizontally without becoming a single point of failure.

SPOTLIGHT Execution Flow. Clients submit updates through Envoy to the RM, where the Scheduler assigns each update to a Sub-Global Aggregator cohort. Sub-Global Aggregators accumulate updates from their assigned clients and produce sub-global models, which Global Aggregators merge into a single consistent global model. Concurrently, the Performance Manager observes load across both aggregation tiers and signals the Scaler to provision or deprovision aggregators as demand shifts. When the Migrator detects elevated failure likelihood on an aggregator, the elected leader directs workload migration to a healthy target and checkpoints the at-risk instance’s current state. If no healthy cloud aggregators are available, a Failover Edge Aggregator assumes the role. The

RM redelivers any in-flight updates to the replacement, and training continues without restarting.

3.2 Three Design Principles

We further elaborate on the three design principles of SPOTLIGHT, explaining how each addresses the insights identified in §??.

Two-tiered aggregation. Scaling aggregators increases coordination and synchronization overhead. SPOTLIGHT manages this through a two-tiered aggregation structure in which each Sub-Global Aggregator handles only its assigned subset of clients, and Global Aggregators receive only a fixed number of sub-global models regardless of total client count. This bounds the coordination scope of each tier independently, allowing both tiers to scale without overhead growing proportionally with the number of clients or aggregators. Global Aggregators then merge the fixed set of sub-global models into a single consistent global model, providing an explicit aggregation step that scales independently of client count.

Proactive fault tolerance. Naive fault tolerance mechanisms that rely on provider termination signals, periodic checkpointing, or restarting lost work are insufficient for FL aggregation on spot infrastructure. SPOTLIGHT instead uses a Kaplan-Meier survival model [22] derived from publicly available spot VM lifetime traces [56] to estimate failure likelihood ahead of preemption, triggering workload migration and state checkpointing before an instance is lost. This ensures that aggregation continues without losing any in-flight updates.

Elastic spot provisioning. On-demand instances make sustained horizontal scaling economically prohibitive. SPOTLIGHT provisions its aggregation layer on spot instances, but the cost advantage of spot can only be realized if the system can absorb aggregator disruptions without stalling throughput. Scalability and fault tolerance are therefore co-equal concerns in SPOTLIGHT, working together simultaneously to sustain aggregator throughput and preserve the cost efficiency of spot infrastructure at scale

4 System Components

Building on the three design principles established in §3.2, this section describes how SPOTLIGHT realizes them in practice. The aggregation tiers realize two-tiered aggregation through independently scalable Sub-Global and Global Aggregators. The Resource Manager handles scheduling, orchestrates scaling across both tiers, and migrates workloads before preemption occurs, preserving training continuity and the cost efficiency of spot infrastructure. We begin with the update lifecycle (§4.1), which establishes the end-to-end path a client update follows through the system, then describe each component in detail.

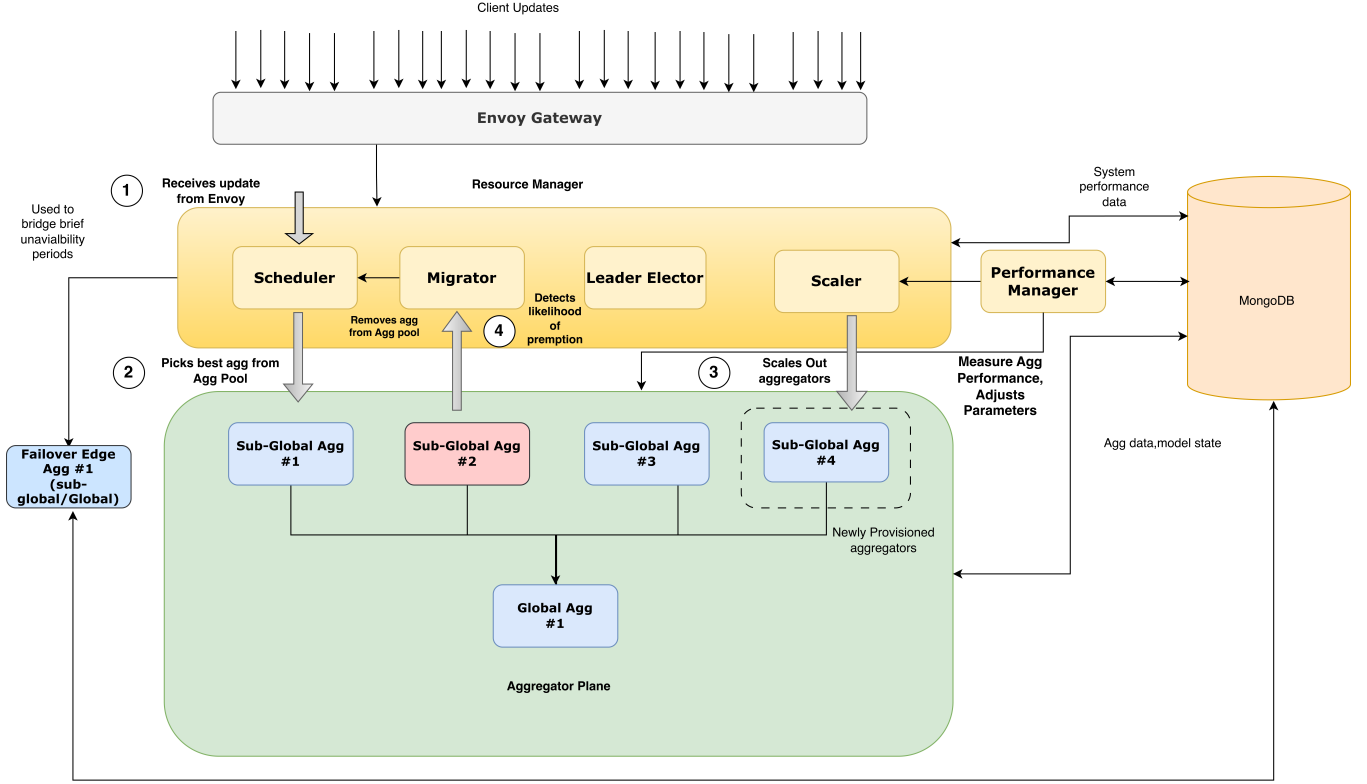


Figure 3: SpotLight architecture. Client updates enter through the Envoy Gateway and are placed by the Scheduler onto Sub-Global Aggregator instances (1, 2). The Scaler provisions new aggregator instances in response to signals from the Performance Manager (3). The Migrator continuously evaluates preemption risk and removes at-risk instances from the Scheduler’s candidate pool (4). Failover Edge Aggregators are utilized temporarily when aggregators are unavailable

4.1 Update Lifecycle

In SPOTLIGHT, client updates flow through two aggregation tiers, accumulating into sub-global models before being merged into the global model. Clients maintain persistent gRPC connections to Envoy, which routes them to the Resource Manager. The RM forwards client updates to the appropriate Sub-Global Aggregator asynchronously in the background, decoupling client-facing latency from aggregator throughput. The Sub-Global Aggregator accumulates updates in an in-memory buffer and triggers aggregation once the floor threshold is met. The resulting sub-global model is pushed to the Global Aggregator, which incorporates it into the global model. Sub-Global Aggregators pull the updated global model and propagate it down the hierarchy to clients.

4.2 Two-Tiered Aggregation

SPOTLIGHT distributes aggregation across two tiers, allowing each tier to scale independently as client participation grows. **Aggregators** operate at two distinct points in the hierarchy with different responsibilities. Sub-Global Aggregators receive updates from the RM, aggregate them into a sub-global

model, and ensure clients receive the latest available model. Global Aggregators receive sub-global models from all Sub-Global Aggregators and merge them into a single global model. Both tiers adopt FedBuff’s [39] buffered aggregation model, but use the buffer as a floor rather than a strict barrier, triggering aggregation as soon as the minimum threshold is met, avoiding stalls from incomplete batches and improving aggregator throughput. Sub-Global Aggregator workload scales directly with client participation and can vary significantly as clients join or leave training, while Global Aggregator workload is bounded by the number of Sub-Global Aggregators, making it more stable. This difference in workload characteristics motivates distinct scaling policies at each tier.

Scaler elastically scales both aggregation tiers in response to client load and system performance. The scaling of Sub-Global and Global Aggregators is managed by the RM. The scaling policy used by SPOTLIGHT for Sub-Global Aggregators utilizes client participation and utilization to scale horizontally, while Global Aggregators use memory utilization to scale. The difference in scaling signal between the two tiers is motivated by the workload characteristics of each tier. The al-

gorithm used for scaling Sub-Global Aggregators is described in Algorithm 2 (Figure 6). The Scaler utilizes this algorithm along with real-time measurements and scaling parameters obtained from the Performance Manager to scale Sub-Global Aggregators. For Global Aggregators, SPOTLIGHT internally implements the Kubernetes HPA algorithm, with scaling parameters likewise supplied by the Performance Manager.

Performance Manager dynamically adjusts the scaling parameters consumed by the Scaler across both aggregation tiers in response to observed system performance. The Performance Manager implements a Proportional-Integral-Derivative (PID) controller that monitors aggregation throughput and reacts to changes between consecutive control cycles. A fixed setpoint is avoided as FL workloads operate under variable client participation, making a stable throughput baseline difficult to establish without extensive tuning. Instead, the controller reacts to changes in throughput between cycles, tightening scaling parameters when throughput degrades and relaxing them when it improves. The error signal $e(t)$ and controller output $u(t)$ are defined in Equations 1 and 2. The PID parameters used in our deployment are listed in Table 1.

Scheduler places incoming client updates onto available Sub-Global Aggregators using a queue-aware placement policy, formally defined in Algorithm-1 (Figure 5) and Figure 4. In SPOTLIGHT, aggregation begins once the minimum number of client updates are present on the aggregator. The bursty nature of FL workloads means naive placement policies risk stalling aggregation by spreading updates thinly across aggregators, none of which reach the floor threshold. The Scheduler addresses this by selecting the two aggregators with the largest queues from the active pool, concentrating updates to trigger aggregation as early as possible. The selected aggregator's AFS is incremented on every scheduling decision and decays over time since the last selection. When aggregators are equally ranked by queue length, AFS serves as a tiebreaker, deprioritizing aggregators with more recent scheduling decisions in favor of those that have been selected less recently. The decision flow is illustrated in Figure 4.

Version Consistency As Sub-Global and Global Aggregators scale independently, concurrent aggregation events across tiers can produce divergent model versions, causing clients to train on stale or inconsistent state. SPOTLIGHT addresses this with a shared monotonic version counter in MongoDB, atomically incremented by all aggregators whenever a new aggregated model is produced. The version counter provides a consistent ordering of aggregation events across the hierarchy, ensuring that clients always receive the most recent model regardless of which aggregator serves it. All aggregators detect and recover stale in-memory state after restarts by comparing their local version against the shared counter, pulling the latest model from persistent storage when a discrepancy is found.

$$e(t) = T_{\text{current}} - T_{\text{previous}} \quad (1)$$

Table 1: PID Controller Values

Parameter	Value
K_p	0.05
K_i	0.02
K_d	0.10
γ	0.97
I_{max}	10.0

$$u(t) = K_p e(t) + K_i \cdot \text{clamp}(\gamma \cdot I_{t-1} + e(t) \cdot \Delta t, -I_{\text{max}}, I_{\text{max}}) + K_d \frac{de(t)}{dt} \quad (2)$$

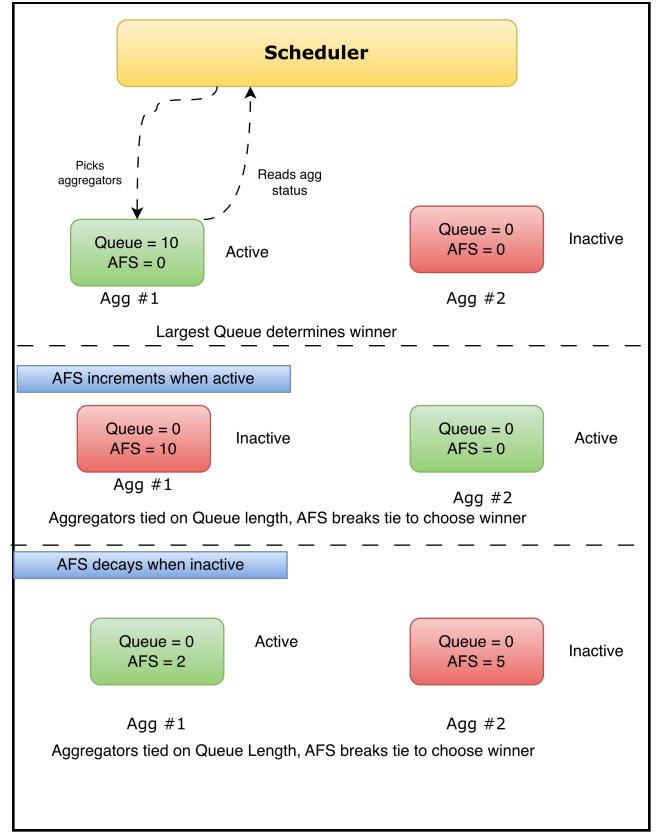


Figure 4: **Scheduling algorithm.** SPOTLIGHT primarily selects the aggregator with the largest queue to begin aggregation; when no aggregator has a clear lead, the Aggregation Fairness Score (AFS) breaks the tie. AFS measures how recently an aggregator was selected, with a higher score indicating heavier recent use. To avoid overloading any single aggregator, the scheduler favors aggregators that have not been picked recently, incrementing AFS each time an aggregator is chosen and decaying it exponentially over time.

Algorithm 1 Scheduling Algorithm

Input: M : metrics $\rightarrow$ aggregator $\mapsto \langle q \rangle$, where q = queue size
Input: AFS: fairness score map (as of last update)
Input: AFS_{ts}: last-update timestamp map
Input: λ : decay factor

```

1: function EFFECTIVEAFS( $p, t$ )
2:   return AFS[ $p$ ]  $\cdot \lambda^{t - \text{AFS}_{ts}[p]}$   $\triangleright$  Continuous decay
3: function ONPICK( $p, t$ )
4:   AFS[ $p$ ]  $\leftarrow$  EFFECTIVEAFS( $p, t$ ) + 1  $\triangleright$  Refresh, then increment
5:   AFSts[ $p$ ]  $\leftarrow t$ 
6:  $C \leftarrow \text{MIGRATOR.GETSCHEDULABLEAGGREGATORS}$   $\triangleright$  Step 1: set pool
7:  $t \leftarrow \text{Now}$ 
8:  $Q_{\max} \leftarrow \max_{p \in C} M[p].q$   $\triangleright$  Step 2: rank by queue
9:  $\mathcal{T} \leftarrow \{p \in C : M[p].q = Q_{\max}\}$   $\triangleright$  Ties
10:  $p^* \leftarrow \arg \min_{p \in \mathcal{T}} \text{EFFECTIVEAFS}(p, t)$   $\triangleright$  Break with AFS
11: ONPICK( $p^*, t$ )  $\triangleright$  Step 3: update AFS
12: schedule on  $p^*$ 

```

Figure 5: Algorithm #1 – Scheduling-pseudo-code – algorithm version of Fig. 4.

Algorithm 1 ScalingAlgorithm

```

// Collect Inputs
1:  $num\_clients \leftarrow \text{GETNUMCLIENTS}$ 
2:  $current\_replicas \leftarrow \text{GETCURRENTREPLICAS}$ 
3:  $scale\_threshold, cooldown\_time \leftarrow \text{GETFROMPERFORMANCE-MANAGER}$ 
4:  $util\_threshold \leftarrow \text{Env.UTIL\_THRESHOLD}$ 

// Step 1: Client-Based Policy
5:  $desired\_replicas \leftarrow \lceil num\_clients / scale\_threshold \rceil$ 
6: if  $desired\_replicas > current\_replicas$  then
7:   Scale out to  $desired\_replicas$ 
8:   BEGINCOOLDOWN( $cooldown\_time$ )
9: else if  $desired\_replicas < current\_replicas$  then
10:  Scale in to  $desired\_replicas$ 
11:  BEGINCOOLDOWN( $cooldown\_time$ )
12: else

// Step 2: Utilization-Based Policy (fallback)
13:  $current\_util \leftarrow training\_time / (training\_time + avg\_wait\_time + avg\_agg\_time)$ 
14: if  $current\_util < util\_threshold$  then
15:    $\lambda_{updates} \leftarrow \text{GETUPDATERATE}$ 
16:    $desired\_replicas \leftarrow \lceil \lambda_{updates} / (scale\_threshold \cdot util\_threshold) \rceil$ 
17:   if  $desired\_replicas > current\_replicas$  then
18:     Scale out to  $desired\_replicas$ 
19:     BEGINCOOLDOWN( $cooldown\_time$ )
20:   else if  $desired\_replicas < current\_replicas$  then
21:     Scale in to  $desired\_replicas$ 
22:     BEGINCOOLDOWN( $cooldown\_time$ )

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Figure 6: Algorithm 2: Scaling Algorithm.

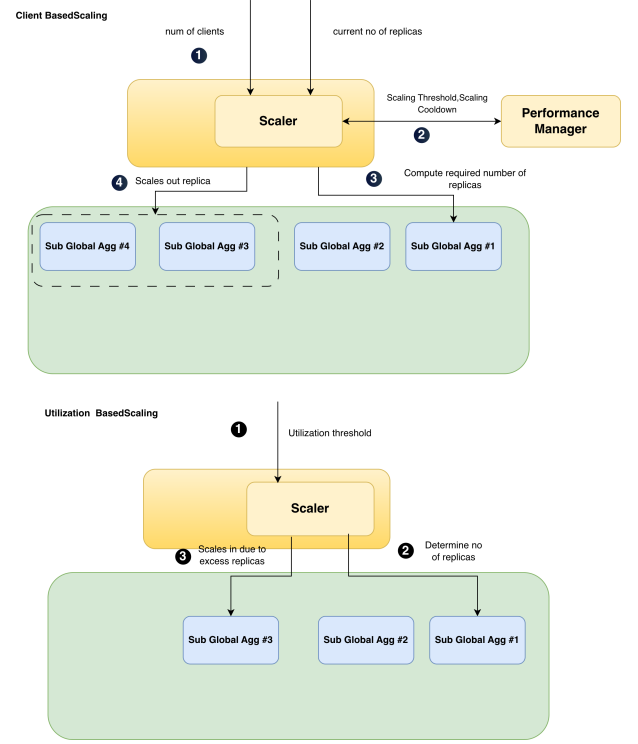


Figure 7: Scaling Algorithm

4.3 Proactive Fault Tolerance

Spot instances can be reclaimed by the cloud provider at any time, making reactive recovery insufficient. SPOTLIGHT utilizes proactive fault tolerance (FT) techniques to mitigate the impact of spot preemptions on training progress.

Migrator is responsible for two key FT operations in SPOTLIGHT: workload migration and checkpoint triggering. The Migrator derives a Kaplan-Meier model [22] from publicly available lifetime traces [56] to estimate the likelihood of aggregator failure based on aggregator age. The Migrator tracks the age and preemption events of aggregators and operates in a closed loop with the Scheduler described in Section 4.2. The Migrator proactively detects aggregators at risk of failure and removes them from the candidate pool. The Scheduler then uses the updated pool to schedule incoming client updates on the available aggregators. When migration is not possible due to aggregator unavailability, the Migrator detects this and directs the Scheduler to trigger a Failover Edge Aggregator, which acts as a temporary bridge until a replacement aggregator is provisioned. The migration mechanism is shown in Fig. 8a and b.

Checkpointing in SPOTLIGHT is triggered by the Migrator when an aggregator is identified as likely to be preempted. The aggregator persists its model state, in-buffer client contributions, and server-side optimizer state to the database. Optimizer state is saved conditionally based on the aggregation algorithm in use: stateless algorithms such as FedAvg require

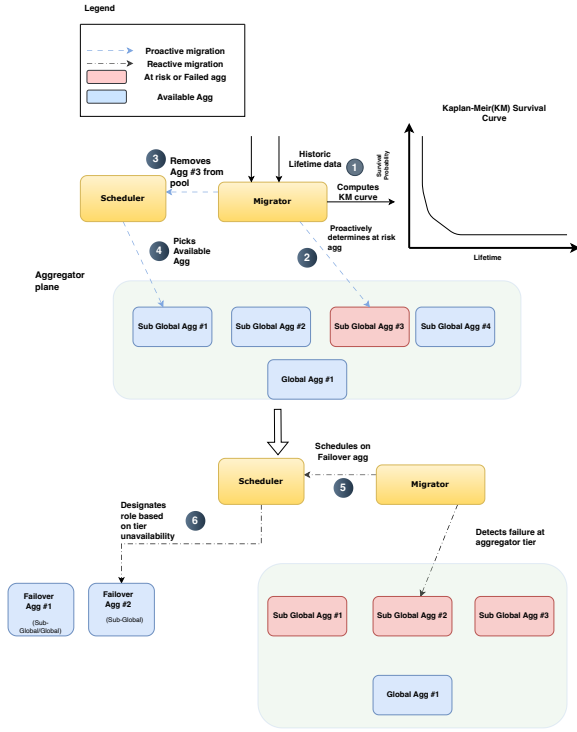


Figure 8: Migrator decision flow. The Migrator uses a Kaplan-Meier survival model [22] to compute survival probability from historic aggregator lifetime data, proactively identifying at-risk aggregators and removing them from the Scheduler’s candidate pool before failure occurs (top). When an aggregator tier fails and no aggregator in the pool is available, the Migrator detects the failure and directs the Scheduler to designate a Failover Aggregator based on which tier became unavailable (bottom).

no optimizer state to be saved, while stateful algorithms such as FedAdam persist their server-side optimizer state alongside the model. This design reduces I/O overhead compared to periodic checkpointing, persisting state only when the Migrator determines recovery is likely to be needed.

Leader Elector protects the RM, which serves as the control plane for SPOTLIGHT, coordinating scaling, migration, and scheduling across the entire system. A single RM instance would represent a critical point of failure; the RM therefore scales horizontally, introducing the need for consensus to prevent conflicting control plane decisions across instances. SPOTLIGHT uses the Raft consensus protocol [40] to maintain a single elected leader among RM instances at all times. The leader holds exclusive authority over scaling and migration decisions, pushing them to all follower instances, which use them as the basis for scheduling incoming client

updates. When the leader fails, Raft elects a successor automatically, restoring full control plane authority without interrupting scheduling operations across instances.

4.4 Elastic Spot Provisioning

SPOTLIGHT addresses two key challenges of FL systems deployed on spot infrastructure, variable client demand and spot preemption, through the coordinated operation of the components described in Sections 4.2 and 4.3. The Scaler and Performance Manager work in a closed loop to maintain aggregation throughput as client participation fluctuates, adjusting provisioning in response to observed system performance. Concurrently, the Migrator and Scheduler operate in a closed loop to handle spot preemptions: the Migrator continuously updates the pool of available aggregators based on predicted preemption risk, while the Scheduler places incoming updates on the aggregators most likely to trigger aggregation soonest within that pool. When no aggregator is available, Failover Edge Aggregators serve as a bridge. Working concurrently, these components allow SPOTLIGHT to adapt to demand variability and infrastructure instability simultaneously, without manual intervention.

5 Implementation

5.1 SPOTLIGHT Implementation

SPOTLIGHT is implemented in Python across 19.2k lines of code, reflecting the breadth of its control plane, fault tolerance, and scheduling components, and is deployed on a Kubernetes cluster. SpotLight utilizes third-party libraries including Motor, an asynchronous MongoDB driver, gRPC for inter-component communication, NumPy for algorithm implementations, and the Kubernetes Python client for cluster management. SpotLight uses MongoDB for coordination, state management, and client lifecycle tracking. System components are deployed as StatefulSets, which provide stable network identities necessary for gRPC-based inter-component communication, and communicate exclusively over gRPC, chosen over REST for its HTTP/2-native multiplexing and bidirectional streaming, which eliminates per-request connection overhead on the critical path of persistent client-to-RM streams. All inbound client connections are routed through an Envoy gateway, selected for its native gRPC support and connection load balancing. SpotLight was hosted on Chameleon Cloud [23] during evaluation. The algorithms supported by SpotLight are listed in Table 2.

Table 2: Supported Aggregation Algorithms

Algorithm	Global Model Update
FedAvg [36]	$w^{(t+1)} = \sum_{k=1}^K \frac{n_k}{N} w_k^{(t)}$
FedBuff [39]	$w^{(t+1)} = \frac{\sum_{k \in \mathcal{B}} n_k w_k^{(t)}}{\sum_{k \in \mathcal{B}} n_k}$
FedNova [53]	$m^{(t+1)} = \beta_m m^{(t)} + (1 - \beta_m) \frac{\sum_k p_k (w^{(t)} - w_k^{(t)})/\tau_k}{\sum_k p_k/\tau_k}, \quad w^{(t+1)} = w^{(t)} - \eta_k m^{(t+1)}$
FedAdam [44]	$w^{(t+1)} = w^{(t)} - \eta \cdot \frac{\hat{m}^{(t)}}{\sqrt{\hat{v}^{(t)} + \epsilon}}, \quad g = w^{(t)} - \tilde{w}$
FedYogi [44]	$w^{(t+1)} = w^{(t)} - \eta \cdot \frac{\hat{m}^{(t)}}{\sqrt{\hat{v}^{(t)} + \epsilon}}, \quad g = \tilde{w} - w^{(t)}$

5.2 Spot Emulator

A key feature of spot instances is their variability in pre-emption behavior, driven by factors including region, availability zone, instance type, temporal market conditions and cloud providers [15, 29, 47]. This variability across configurations makes real-world evaluation of application behavior on spot instances difficult to reproduce consistently and cost-prohibitive at scale, as comprehensive coverage across regions and instance types requires extensive testing for long periods of time. Therefore, to evaluate SpotLight under realistic spot conditions, we developed the Spot Emulator, a high-fidelity trace-driven emulator that replays spot VM behavior using publicly available spot availability traces [27, 56]. The emulator is designed around two key goals. First, high fidelity: the emulator replays preemption events derived from real-world availability traces, ensuring emulated behavior reflects realistic spot market conditions. Second, controller-agnostic design: the modular architecture of the Spot Emulator allows it to support Spot VM behavior replay with different cloud services providers and deploy on all platforms with minimal changes to the emulator.

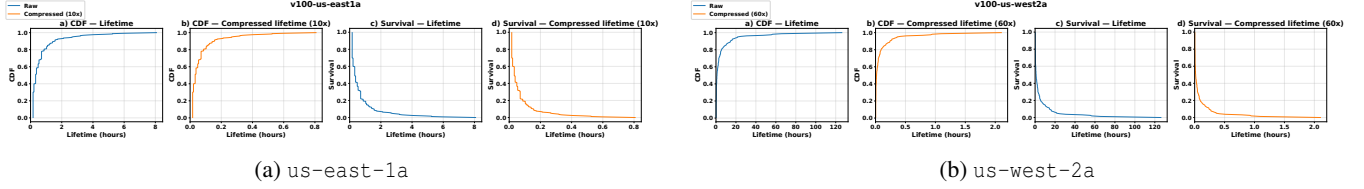


Figure 9: Lifetime and survival distributions for V100 spot instances, derived from availability traces in [56]. Panels (a) and (b) in each subfigure show the CDF of instance lifetime in raw and time-compressed form; panels (c) and (d) show the corresponding Kaplan-Meier survival curve. The time compression factor ($10\times$ for us-east-1a, $60\times$ for us-west-2a) is a configurable parameter of the Spot Emulator, set independently per region to fit evaluation run-time constraints; it is not derived from the trace data itself. The Spot Emulator samples instance lifetimes from these distributions to replay preemption events during evaluation.

The lifetime distribution used to replay spot behavior is derived from the availability traces in [56]. The availability trace is used to derive a CDF, which allows the emulator to reliably replay spot instance lifetime dynamics. To ensure that evaluation cost is realistic, price traces obtained from [27] are used to generate realistic execution costs. As shown in Figure 9, survival probability decreases monotonically with instance lifetime, reflecting the inherent unreliability of spot instances over time.

The Spot Emulator consists of five components. The first is the *Distribution Extractor*, which filters the availability trace by region, availability zone, and OS, and derives the lifetime CDF from the filtered data. The second is the *Lifecycle Manager*, which samples a lifetime for each newly provisioned VM from the CDF within a configured range, excluding tail outliers that represent atypically short-lived or long-lived instances, and maintains an internal clock tracking the lifetime of all active spot VMs. The third component is the *Executor*, which terminates instances via the Kubernetes API once their assigned lifetime expires. The fourth component is the *Control Plane*, which manages the execution order of the other components and exposes configuration parameters to the user, including the sampling range and time compression factor. The fifth component is the *Cost Report Generator*, which uses price traces from [27] to produce accurate execution cost reports.

Execution begins when the Control Plane is configured with the lifetime trace source, sampling range, and time compression factor. The Control Plane invokes the Distribution Extractor to generate the lifetime CDF, which is passed to the Lifecycle Manager. The Lifecycle Manager samples a lifetime from the CDF within the configured range, assigns it to each pod, and tracks elapsed time via an internal clock. When a pod’s assigned lifetime elapses, the Lifecycle Manager invokes the Executor to terminate the pod.

The Spot Emulator’s logical model maps directly onto a Kubernetes cluster. The Cluster represents the datacenter, and the pods scheduled onto its worker nodes represent the Spot VMs, each assigned a lifetime and terminated once that lifetime

expires. SpotLight is deployed on top of this emulated cluster, allowing it to be evaluated under realistic spot preemption behavior.

6 Evaluation Setup

SpotLight was evaluated on Chameleon Cloud [23], a platform designed for experimental systems research. SpotLight was deployed on a five-node Kubernetes cluster, with node specifications listed in Table 3. MongoDB was hosted on a dedicated machine outside the cluster, with equivalent specifications to the host node. The evaluation focuses on two key aspects of the system: scalability and fault tolerance.

Table 3: Cluster Node Specifications

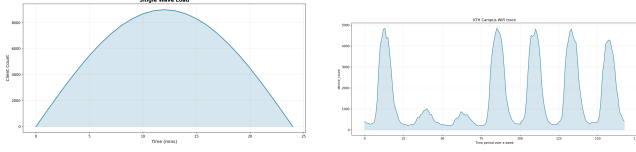
Specification	Host	Worker
CPU	Intel Xeon Gold 6126 @ 2.60GHz	—
vCPUs	48	16
Total RAM	187 GiB	28 GiB

6.1 Scalability Results

SpotLight is evaluated against Papaya [17], a production asynchronous FL system, as the baseline, chosen for its support of asynchronous aggregation at scale.

In order to emulate the behavior of FL clients, a testing platform was designed that follows the typical lifecycle of an FL client from registration to exit. The testing platform has configurable parameters listed in Table 4; the parameters allow the user fine-grained control over the behavior of the clients. The emulated client platform allows the user to emulate a large client population as well as the behavior of a typical FL client. The workload arrival patterns shown in Fig 10 are supported by our platform currently and can be easily added as well.

The scalability experiments use the parameters listed in Table 5 to load test the aggregators. The parameters were chosen to keep per-client execution time small, allowing ex-



(a) Single-wave synthetic trace. (b) KTH campus WiFi trace [41].

Figure 10: Client arrival patterns used in the evaluation. The synthetic trace provides a controlled load shape for isolating scalability behavior, while the KTH trace captures the real-world variation in active client count over the course of a week, used to validate that the cost gains observed under the synthetic trace hold under realistic client arrival patterns.

periments to focus mainly on aggregator scalability without getting bottlenecked by client side limitations.

Table 4: Emulated Client Configuration Parameters

Parameter	Description
Trace	Determines the number of clients active at any given time, governing the client population dynamics
No. of rounds	Number of training rounds the system is configured to run
Training time	Duration of local training at each client
Model size	Size of the FL model being trained

Table 5: Client Configuration for Scalability Experiments

Parameter	Value
No. of rounds	2
Training time	2s
Model size	10MB

The scalability experiments use the parameters listed below to measure the impact of scaling on aggregator performance.

- 1) Client Utilization.** We define client utilization as $\frac{T_{train}}{T_{train} + T_{aggregation} + T_{queue}}$, where T_{train} denotes local computation at the client, $T_{aggregation}$ denotes the time spent waiting for aggregation to complete, and T_{queue} denotes the time buffered at the Sub-Global Aggregator prior to aggregation. Higher utilization indicates a greater fraction of total time spent on useful local computation rather than coordination overhead.
- 2) No of aggregators** This metric evaluates how the aggregation tier scales relative to client load across all supported algorithms. A scalable design should demonstrate proportional and controlled growth in Sub-Global Aggregator instances as client population increases.
- 3) Queue Time.** Queue time measures the duration each client spends waiting before its update enters aggregation, capturing contention at the aggregation layer. In a well-scaled system, queue time should remain bounded as client count grows.

4) Cost. Cost measures the total infrastructure expenditure required to operate SPOTLIGHT under varying client loads, expressed as instance-hours consumed by the aggregation tier over the duration of training.

5) Aggregator Throughput. Aggregator throughput measures the rate at which the aggregation tier processes client updates, expressed as updates aggregated per unit time.

6) Cost per Aggregation. Cost per aggregation measures the infrastructure expenditure incurred per completed aggregation event, computed as total instance-hour cost divided by the number of aggregations performed over the same run. This metric isolates whether cost efficiency holds as a function of work performed.

The scalability evaluation utilizes the two workload arrival patterns shown in Figure 10. The single-wave pattern provides a controlled, steadily increasing load, isolating scalability behavior from real-world variation. The KTH WiFi trace [41] evaluates the system under realistic conditions, where the number of active clients varies due to diurnal patterns, validating that gains observed under the single-wave pattern hold under real-world client arrival behavior.

Papaya is not scalable by design therefore in order to evaluate the performance of Papaya against Spotlight fairly, the configurations of Papaya are determined below

1) Papaya-Budget. The number of aggregators is derived from SpotLight’s execution cost, comparing performance under an equal budget for both systems.

2) Papaya-Max. Papaya’s aggregators are scaled up until its client utilization and queue time approach SpotLight’s, quantifying the additional cost required to match SpotLight’s performance without elastic scaling.

SPOTLIGHT is first evaluated using the single-wave workload with the configurations above to establish baseline scalability behavior. The KTH trace is then used to evaluate the framework under realistic client arrival conditions, ensuring SPOTLIGHT is tested comprehensively across both controlled and real-world settings.

6.1.1 Single-Wave Workload

This subsection tests whether SpotLight’s scaling behavior holds despite the varying per-round overhead across aggregation algorithms, under the controlled, single-wave synthetic trace shown in Fig 10a, evaluated across FedAvg, FedAdam, FedNova, and FedYogi.

Client Utilization. As summarized in Table 6, SpotLight sustains a median utilization of 0.98 across all four algorithms (Fig 11a, 12a, 13a, 14a). Papaya-Budget is similarly stable but at a lower ceiling, holding a median of 0.80 in all four cases (Fig 11e, 12e, 13e, 14e); this flat curve indicates a fixed capacity limit rather than a bottleneck that worsens with client count. Papaya-Max varies more by algorithm, holding a median of 0.95 under FedAvg and FedNova, 0.93 under FedAdam, and 0.94 under FedYogi.

Table 6: Median client utilization across all four aggregation algorithms under the single-wave trace. SL = SpotLight.

Algorithm	SL	PBudget	PMax
FedAvg	0.98	0.95	0.96
FedAdam	0.98	0.90	0.96
FedNova	0.98	0.93	0.96
FedYogi	0.98	0.94	0.94

Table 7: Average queue time across all four aggregation algorithms under the single-wave trace. SL = SpotLight.

Algorithm	SL	PBudget	PMax
FedAvg	0.67s	1.54s	0.71s
FedAdam	0.67s	7.61s	0.88s
FedNova	0.70s	1.81s	0.71s
FedYogi	0.67s	1.03s	1.05s

Client Utilization. As summarized in Table 6, SpotLight sustains a median utilization of 0.98 across all four algorithms (Fig 11a, 12a, 13a, 14a). Papaya-Budget is similarly stable but at a lower ceiling, holding a median of 0.80 in all four cases (Fig 11e, 12e, 13e, 14e); this flat curve indicates a fixed capacity limit rather than a bottleneck that worsens with client count. Papaya-Max varies more by algorithm, holding a median of 0.95 under FedAvg and FedNova, 0.93 under FedAdam, and 0.94 under FedYogi.

Queue Time. As summarized in Table 7, SpotLight’s average queue time stays between 0.64s and 0.67s across all four algorithms (Fig 11b, 12b, 13b, 14b), with a tightly clustered distribution in each case (Fig 11c, 12c, 13c, 14c). Papaya-Budget is consistently the slowest of the three, averaging between 3.77s and 3.78s in every algorithm, with almost no distribution overlap against the other two systems. Papaya-Max averages 0.71s under both FedAvg and FedNova, 0.87s under FedAdam, and 1.04s under FedYogi, the widest gap to SpotLight of the four algorithms, with a distribution tail extending to 4.1s.

Cost. SpotLight and Papaya-Budget are held to equal cost by construction in every algorithm (Fig 11d, 12d, 13d, 14d), so the cost comparison reduces to how much more Papaya-Max must spend to approach SpotLight’s utilization and queue time. As shown in Table 8, Papaya-Max costs \$22.70 under FedAvg against SpotLight’s \$14.43, and \$22.70 under FedNova against SpotLight’s \$13.90, both bringing it reasonably close to SpotLight’s utilization and queue time. Under FedAdam and FedYogi, Papaya-Max costs \$22.44 and \$22.70, respectively, against SpotLight’s \$11.99 in both cases, yet still falls short of SpotLight on both metrics. Across all four algorithms, matching SpotLight’s performance without elastic scaling requires Papaya-Max to spend nearly double SpotLight’s cost or more, and that additional spend does not fully close the gap under FedAdam or FedYogi.

Table 8: Total infrastructure cost across all four aggregation algorithms under the single-wave trace. SL = SpotLight. Papaya-Budget is provisioned at equal cost to SpotLight in every case.

Algorithm	SL Cost	PBudget Cost	PMax Cost
FedAvg	\$14.36	\$14.36	\$19.64
FedAdam	\$7.62	\$7.62	\$11.48
FedNova	\$10.25	\$10.25	\$13.52
FedYogi	\$7.62	\$7.62	\$11.48

6.1.2 KTH Campus WIFI Trace

The KTH campus wifi trace is utilized in order evaluate the SPOTLIGHT under real world conditions. The scalability of SPOTLIGHT with the KTH trace is evaluated with 2 algorithms, the first being FedAvg for stateless and FedAdam for stateful algorithms. Papaya is configured with the 5 aggregator, the same configuration that was followed with Papaya-Max in the previous section 6.1.1.

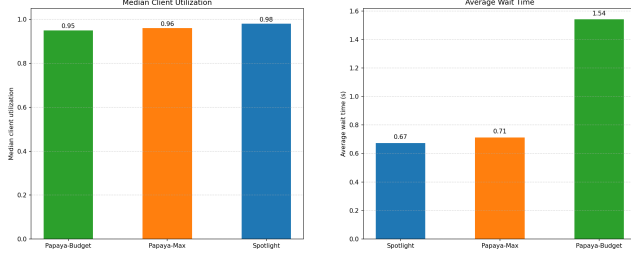
FedAvg

FedAdam

Across both aggregation algorithms, SPOTLIGHT outperforms Papaya on cost, cost per aggregation, median throughput, and client utilization on the KTH trace.

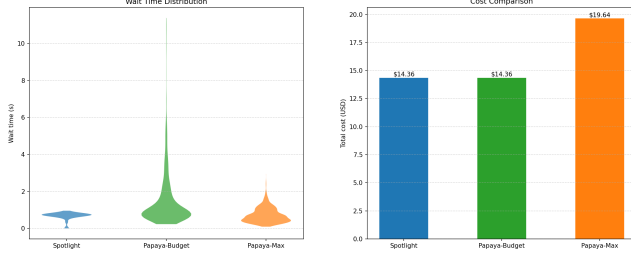
FedAvg. Figure 15 shows SPOTLIGHT’s replica count tracking client demand over the trace. Over a projected 168-hour run (Figure 17), SPOTLIGHT costs \$1,298.68 compared to \$2,570.40 for Papaya, a reduction of approximately 49.5%. This lower cost comes without sacrificing performance: as shown in Figure 18, SPOTLIGHT sustains higher throughput than Papaya throughout the trace, with a median of 57.40 updates/sec against Papaya’s 48.26 (Figure 19), roughly 1.19x higher, while spending only \$0.11 per aggregation versus Papaya’s \$0.24 (Figure 20), a 2.18x reduction. Client utilization follows the same pattern (Figure 16): SPOTLIGHT remains consistently above 0.97, while Papaya fluctuates more, dipping as low as approximately 0.95 at several points during the trace.

FedAdam. The same pattern holds under FedAdam, though the balance shifts toward throughput. Figure 21 shows SPOTLIGHT’s replica count over the trace, never dropping below two. SPOTLIGHT’s projected 168-hour cost is \$2,010.40 against Papaya’s \$2,570.40 (Figure 23), a 21.8% reduction, while Figure 24 shows SPOTLIGHT sustaining a wider throughput advantage than under FedAvg, with median throughput reaching 97.78 updates/sec compared to Papaya’s 48.26 (Figure 25), roughly 2.03x higher. Cost per aggregation falls to \$0.05 for SPOTLIGHT versus \$0.24 for Papaya (Figure 26), a 4.8x reduction. Client utilization under FedAdam is notably stable for SPOTLIGHT, holding essentially flat at approximately 0.99 for the full duration shown (Figure 22), while Papaya’s utilization continues to fluctuate between roughly 0.95 and 0.98.



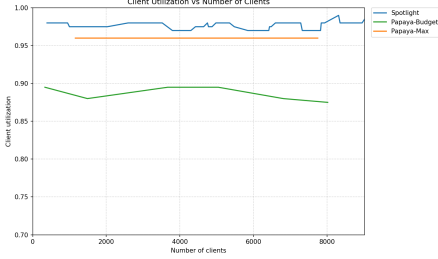
(a) Median client utilization.

(b) Average queue time.



(c) Queue time distribution.

(d) Total cost comparison.

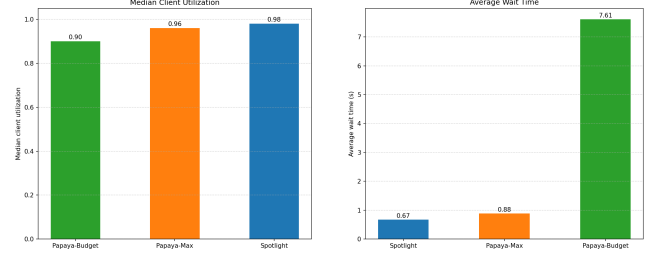


(e) Client utilization vs. number of clients.

Figure 11: FedAvg results: client utilization, queue time, and cost under the single-wave trace.

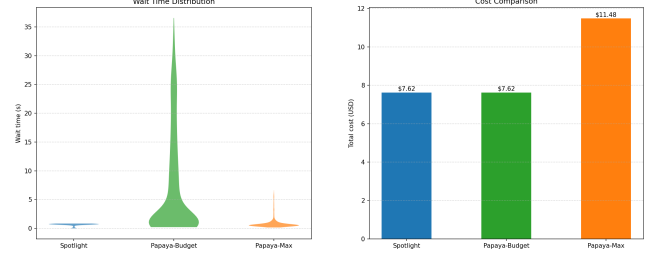
SPOTLIGHT outperforms Papaya consistently across all metrics. The performance and cost efficiency gains between SPOTLIGHT and Papaya are driven by two key components in the system: the Scaler and the Scheduler. An even distribution of updates leads to queuing delays under the bursty nature of FL workloads, since it assigns updates without regard to which aggregators are closer to triggering aggregation. SPOTLIGHT’s load-aware scheduling algorithm instead uses aggregator state to place incoming updates on the aggregators most likely to begin aggregation soonest, reducing these delays, consistent with the wait-time results in Section 6.1.1. The Scaler, meanwhile, adds aggregators in response to client load, while the Scheduler exploits that added capacity to sustain high throughput. Together, the Scaler and Scheduler allow SPOTLIGHT to meet quickly client demand while keeping overall execution cost lower than Papaya configured for peak load under the max configuration.

While SPOTLIGHT outperforms Papaya under both algorithms, the magnitude of that advantage, particularly in over-



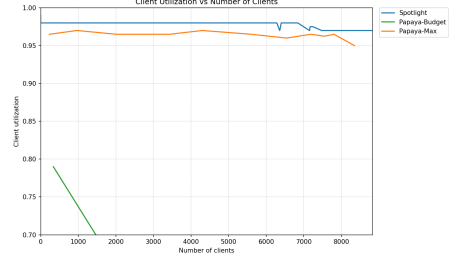
(a) Median client utilization.

(b) Average queue time.



(c) Queue time distribution.

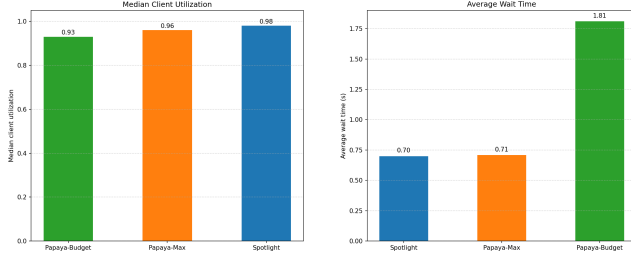
(d) Total cost comparison.



(e) Client utilization vs. number of clients.

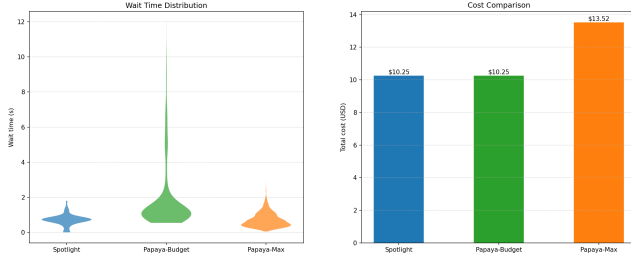
Figure 12: FedAdam results: client utilization, queue time, and cost under the single-wave trace.

all cost, differs between them: running SPOTLIGHT with FedAdam costs 54.8% more than with FedAvg for the same workload, a gap driven by FedAdam’s stateful design. Because the aggregator must retain state throughout the aggregation process, memory contention becomes a bottleneck: during evaluation, this contention caused timeouts and other performance issues when FedAdam ran with as few as one aggregator. SPOTLIGHT was therefore configured with a minimum of two aggregators for FedAdam, in contrast to the single-aggregator minimum used elsewhere. This shift is reflected across the entire trace, not just at the floor: as shown in Figure 27, FedAvg spends 96% of its runtime at one or two replicas, while FedAdam spends 59% of its runtime at three or four, a difference directly reflected in the cost gap between the two algorithms.



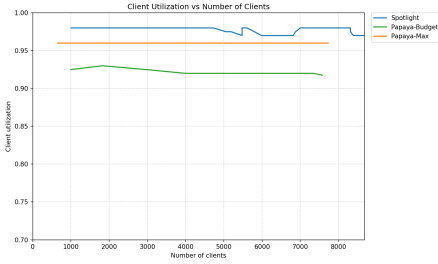
(a) Median client utilization.

(b) Average queue time.



(c) Queue time distribution.

(d) Total cost comparison.



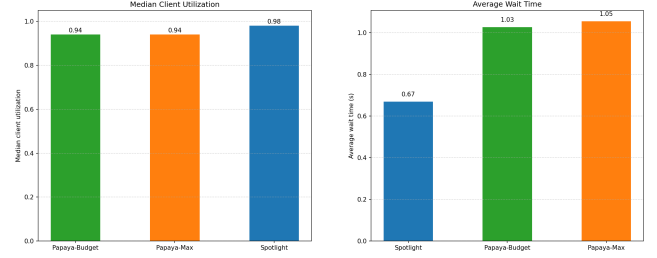
(e) Client utilization vs. number of clients.

Figure 13: FedNova results: client utilization, queue time, and cost under the single-wave trace.

6.2 Fault Tolerance

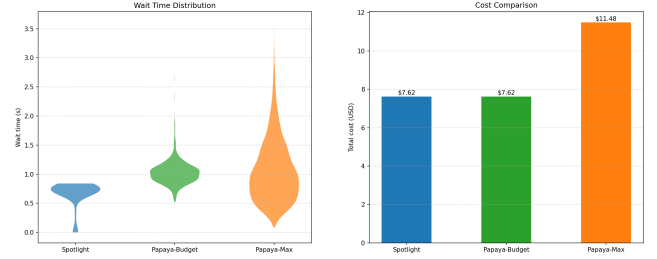
To evaluate the fault tolerance of SpotLight, an ablation study was conducted on CIFAR-10 [25], training 10 clients under both IID and non-IID data distributions, with the non-IID split generated using a Dirichlet distribution with $\alpha = 1.2$. The Spot Emulator described in Sec 5.2 was used to inject Spot VM preemption while the FL process was occurring. The fault tolerance policy was varied across runs in order to isolate the individual strengths of each policy. The configurations are described later in this section.

Configurations. Table 10 summarizes the ten configurations evaluated. **Spotlight-UB** is an upper-bound configuration in which no faults were injected. **Spotlight-LB** is a lower-bound configuration in which fault tolerance was disabled entirely, so faults were injected but never recovered from. **Naive-CP** checkpoints the model whenever a checkpoint is available, without any coordination on timing or placement. **Naive-migration** migrates the aggregator’s state to a replacement node only after the aggregator has already



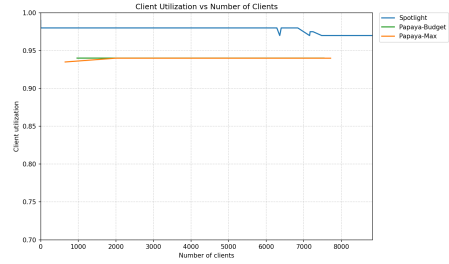
(a) Median client utilization.

(b) Average queue time.



(c) Queue time distribution.

(d) Total cost comparison.



(e) Client utilization vs. number of clients.

Figure 14: FedYogi results: client utilization, queue time, and cost under the single-wave trace.

failed. **Naive-migration+CP** combines naive checkpointing and naive migration, without failover aggregator offloading. **Naive-Full** extends Naive-migration+CP with failover aggregator offloading, so that checkpointing, migration, and offloading are all decided naively. **KM-CP**, **KM-migration**, and **KM-migration+CP** apply SpotLight’s KM configuration to checkpointing, migration, or both, respectively, without failover aggregator offloading. **KM-Full** is SpotLight’s default configuration, in which the KM configuration governs checkpointing, migration, and failover aggregator offloading together.

In IID and non-IID configurations, KM-Full reaches 87.88% and 76.05% respectively, the closest of any fault-tolerant configuration to the fault-free upper bound in each setting (88.97% and 81.00%), outperforming the next-best configuration in each case, Naive-Full (74.64%) under IID and KM-migration+CP (73.37%) under non-IID, by 13.2 and 2.7 percentage points respectively. In both settings, shown in Fig 30 and Fig 31, KM-Full is the best-performing con-

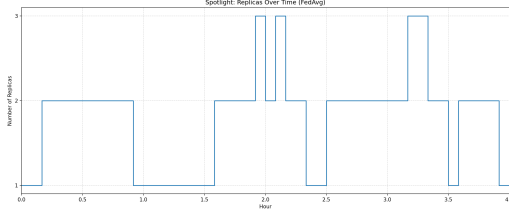


Figure 15: Number of SpotLight replicas over time under FedAvg, tracking the client trace shown in Figure 10b.

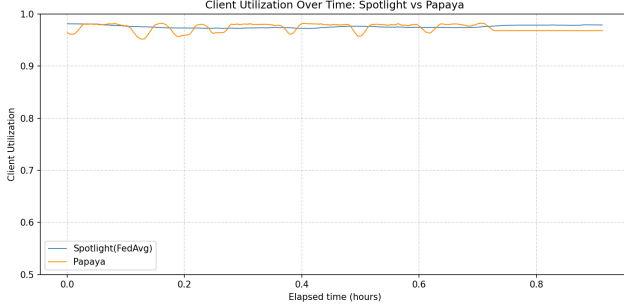


Figure 16: Client utilization over time, SpotLight vs. Papaya, on the KTH trace under FedAvg.

figuration among those that recover from Spot preemptions.

KM-Full’s advantage over the other configurations stems from the way SPOTLIGHT’s three fault tolerance mechanisms reinforce one another. The Migrator anticipates failure and relocates at-risk aggregator instances before they go down, rather than waiting to react. Checkpointing follows the same logic: state is saved only when needed, keeping overhead low while still producing richer checkpoints than Naive-CP, which checkpoints every time a new global model is formed. When migration is not possible, because no cloud aggregator remains available, failover aggregators absorb the load to ensure FL process does not stall.

Figure 32 compares checkpointing behavior between the KM configuration and Naive-CP. The KM configuration takes 83 checkpoints at an average of 784ms each, while Naive-CP takes 159 checkpoints at an average of 1706ms each, roughly half as many checkpoints at less than half the average checkpoint time. This reduction follows from SpotLight’s design: the aggregator serves the global model to clients from an in-memory copy, synchronized with the model stored in the database in the background. The KM configuration instructs the aggregator to checkpoint only when it is at risk of imminent failure, rather than saving at every instance of a new global model, reducing checkpointing overhead. Despite taking fewer checkpoints, the KM configuration still achieves higher accuracy than Naive-CP in both the IID and non-IID settings, as shown in Figures 30 and 31. This indicates that the KM configuration’s reduction in checkpointing overhead

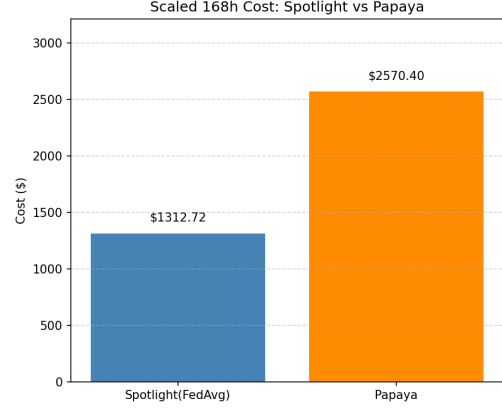


Figure 17: Projected total cost of SpotLight vs. Papaya over 168 hours (one week) on the KTH trace under FedAvg.

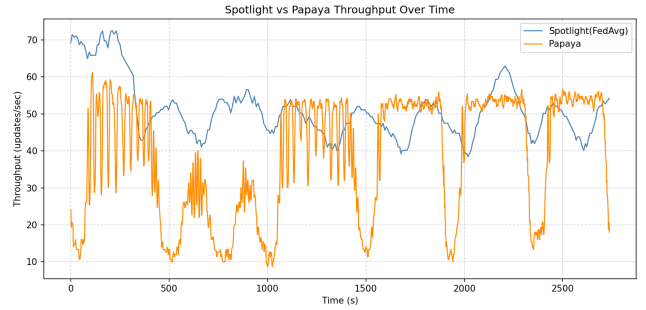


Figure 18: SpotLight vs. Papaya throughput over time on the KTH trace under FedAvg.

does not come at the cost of recovery quality.

Proactively migrating aggregator state ahead of predicted failure, rather than only after failure occurs, allows KM-Full to retain updates that reactive migration would otherwise lose. Figure 33 shows this directly: KM-Full achieves a 99.6% retention rate, compared to 89.4% for NM-Full. Failover aggregators provide a bridge during periods of aggregator unavailability when migration alone is not possible. Together, these mechanisms, proactive migration, low-overhead checkpointing, and failover aggregation, allow KM-Full to outperform the other baselines evaluated in this section.

6.3 Fault Tolerance Under Scaling

Section 6.2 evaluates SPOTLIGHT’s performance while training FL clients, measuring the degradation of model performance under Spot preemption across different configurations of fault tolerance mechanisms. This section evaluates aggregator performance under Spot preemption, to understand how the fault tolerance mechanisms discussed in Section 6.2 and the scaling mechanisms discussed in Section 6.1 work together to sustain throughput and client utilization at scale.

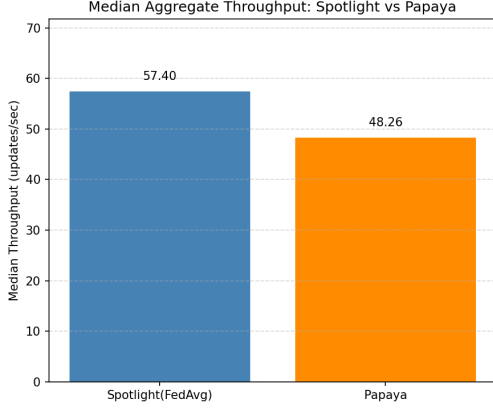


Figure 19: Median aggregate throughput, SpotLight vs. Papaya, on the KTH trace under FedAvg.

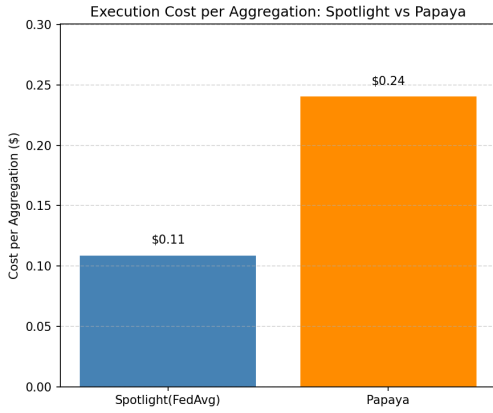


Figure 20: Execution cost per aggregation, SpotLight vs. Papaya, on the KTH trace under FedAvg.

The evaluation in the sections are conducted using the KTH trace in Fig 10b.

Section 6.2 showed that SPOTLIGHT’s full KM policy achieves the highest accuracy among the evaluated fault tolerance configurations. Figure 34 quantifies one source of that gain at scale: the number of client updates received and successfully processed under three FT mechanisms, migration disabled, migration enabled without failover aggregators, and migration enabled with failover aggregators. The full policy, with both migration and failover aggregators enabled, performs best of the three, aggregating approximately 95.3% of updates received under the KTH workload. The full policy with offloading enabled ensures that updates from the clients are aggregated by the system and not lost due to aggregator unavailability.

The evaluation of SPOTLIGHT for FT under scaling begins with the Emulator configured to use the V100 lifetime trace, compressed by 10x, to drive Spot preemption, visualized in Fig 9a. This configuration is referred to in the results below

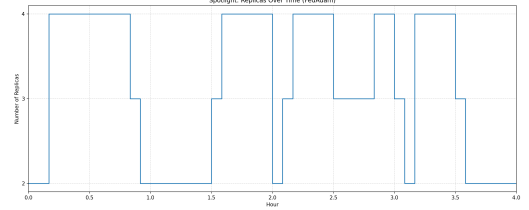


Figure 21: Number of SpotLight replicas over time under FedAdam, tracking the client trace shown in Figure 10b.

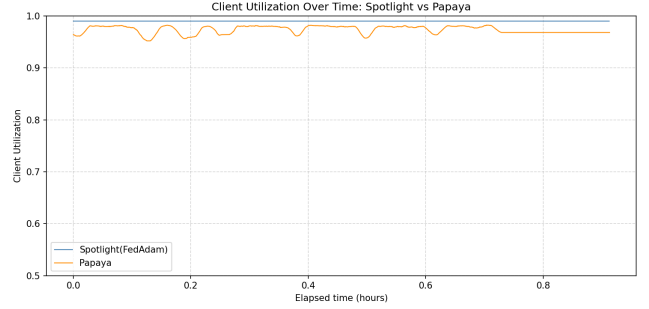


Figure 22: Client utilization over time, SpotLight vs. Papaya, on the KTH trace under FedAdam.

as the base configuration. Figure 35 shows the results for the base configuration. Available indicates periods when cloud aggregators are up and receiving client updates, while unavailable indicates periods when cloud aggregators are down and SPOTLIGHT offloads aggregation to failover aggregators. The top panel of Fig 35 shows the Scaler evaluated in Section 6.1, operating simultaneously with SPOTLIGHT’s fault tolerance mechanisms: replica count rises and falls in response to client demand, the same behavior shown fault-free in Section 6.1.2, while aggregator throughput and client utilization drop when cloud aggregators become unavailable and recover once they become available again. This demonstrates that SPOTLIGHT’s scaling and fault tolerance mechanisms operate together, not independently: the system continues to scale to meet client demand while simultaneously absorbing aggregator failures.

Under the base configuration, SPOTLIGHT spends 88% of its overall execution time with cloud aggregators available, and the remaining 12% offloading to failover aggregators when cloud aggregators are unavailable. The system absorbs failures and utilizes failover aggregators to ensure that the FL process is not disrupted.

While the results above show that SPOTLIGHT can handle the base configuration, the frequency of failures is increased for the same lifetime trace to further test SPOTLIGHT’s performance under increased preemption. The configurations used are listed in Table 9 and visualized in Fig 37.

The medium and high configuration increase the failures rate causing SPOTLIGHT to rely more on failover aggregator to bridge periods of unavailability. The results for medium and

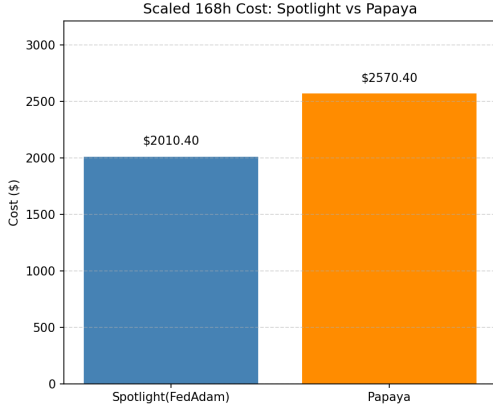


Figure 23: Projected total cost of SpotLight vs. Papaya over 168 hours (one week) on the KTH trace under FedAdam.

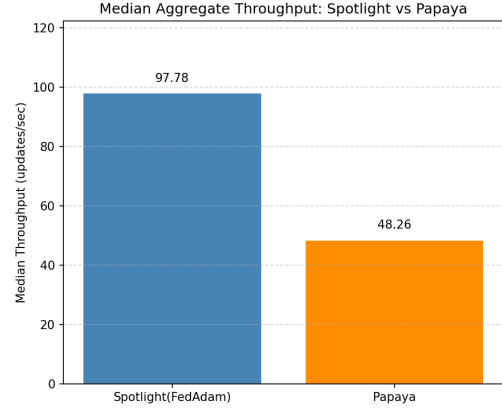


Figure 25: Median aggregate throughput, SpotLight vs. Papaya, on the KTH trace under FedAdam.

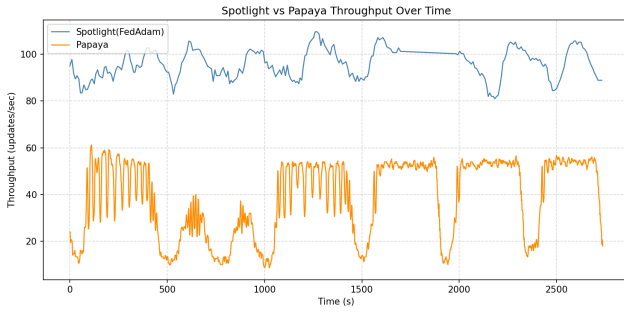


Figure 24: SpotLight vs. Papaya throughput over time on the KTH trace under FedAdam.

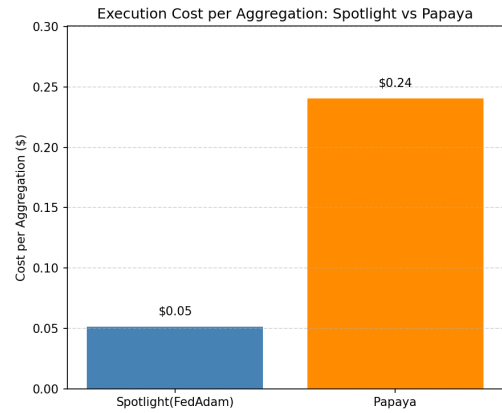


Figure 26: Execution cost per aggregation, SpotLight vs. Papaya, on the KTH trace under FedAdam.

high configuration are seen in Fig 38 and Fig 39.

As SPOTLIGHT adapts to increased failure by relying more heavily on failover aggregators, the share of time spent unavailable rises from 12% at base, to 15% at medium, to 67% at high, shown in Fig 40. Fig 41 shows the corresponding effect on aggregator throughput and client utilization with increased failure. In the case of aggregator throughput, the time spent on cloud aggregators decreases with increased failure, leading to reduced throughput. The throughput on failover aggregators is restricted due to resource constraints and does not match the throughput of the cloud aggregators in any of the three configurations. Client utilization also declines with increased failure, but by a much smaller margin than throughput does, indicating that clients are not forced to wait substantially longer despite the decreasing availability of cloud aggregators.

6.3.1 Recovery from Failure

The results above show that SPOTLIGHT uses migration and offloading to remain available during failure. A fault tolerant system must also recover from failure, not just tolerate it.

To measure SPOTLIGHT’s ability to recover from an outage, we measure whether aggregator throughput returns to at least 70% of its pre-outage baseline, sustained for three consecutive samples, before the next outage begins.

Under this definition, 77% of outages recover at the base configuration, falling to 74% at medium and 38% at high. This indicates that SPOTLIGHT increasingly has less opportunity to stabilize between failures as the failure rate increases: at the high configuration, most outages are followed by another before throughput has a chance to return to its pre-outage level.

This drop in recovery rate tracks a corresponding drop in total available time: SPOTLIGHT spends 9,050 s available at the base configuration, 7,456 s at medium, and 2,475 s at high, shown alongside recovery rate in Fig 43. At the high configuration, the bulk of time is spent on failover aggregators, which, due to resource constraints, cannot match the throughput of the cloud aggregators, leaving outages proportionally less time to recover before the next one begins.

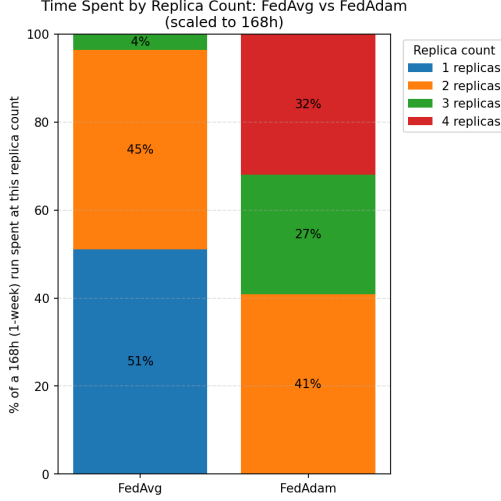


Figure 27: Percentage of a 168-hour (1-week) run spent at each replica count, SPOTLIGHT under FedAvg vs. FedAdam.

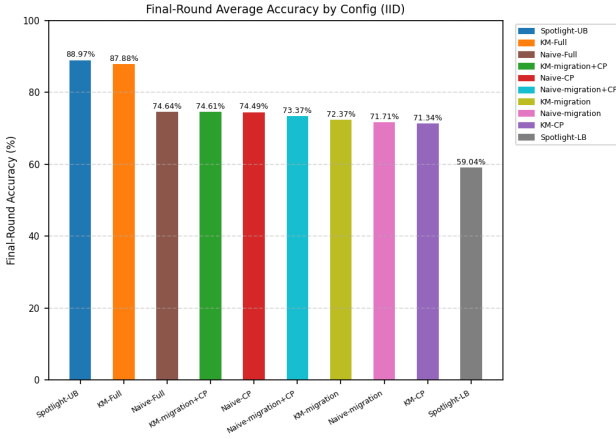


Figure 28: Average accuracy across fault tolerance configurations under IID client data (10 clients).

6.4 SpotLight vs Others

SPOTLIGHT was further evaluated against two existing fault-tolerant FL systems, Totoro+ [11] and FedGuard [5], under the IID CIFAR-10 setting [25] described in §6.2. All three systems were subjected to the same preemption pattern, using the Spot Emulator’s base configuration (the us-east1a-v100 trace at 10x compression, described in §5.2), while each system used its own recovery mechanism to respond: SPOTLIGHT ran under its default KM-Full configuration, Totoro+ ran under its neighborhood-replication recovery mechanism, and FedGuard ran under its own leader-election and checkpointing mechanism. Each run continued until accuracy plateaued, at which point recording stopped.

Figure 44 reports mean accuracy for each system. SPOTLIGHT reaches 87.88%, compared to 39.57% for FedGuard

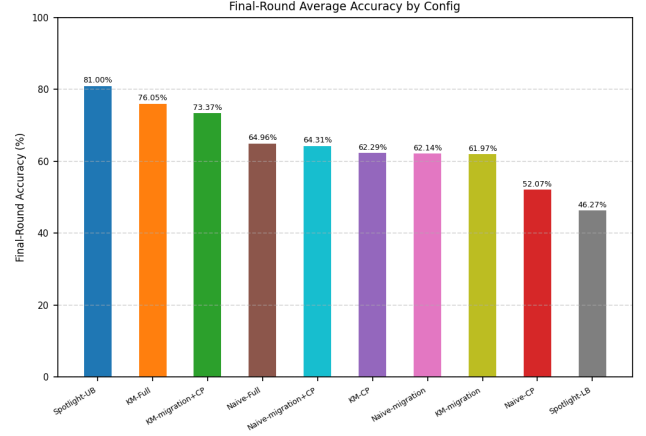


Figure 29: Average accuracy across fault tolerance configurations under non-IID client data (10 clients, Dirichlet $\alpha = 1.5$).

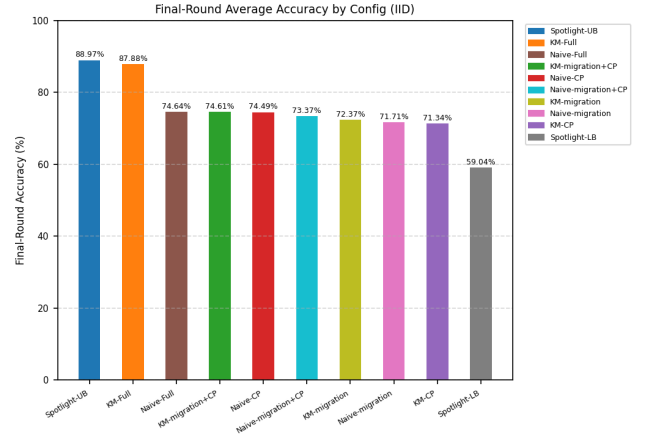


Figure 30: Average accuracy across fault tolerance configurations under IID client data (10 clients).

and 21.73% for Totoro+. Figure 45 reports median downtime in seconds across the failover events encountered during each run. SPOTLIGHT’s downtime is defined as the average time spent in failover, since SPOTLIGHT is designed so that client updates continue to be accepted during recovery and does not, strictly speaking, incur downtime the way a single-aggregator system does; this metric was substituted so that all three systems could be compared on a common axis. Under this definition, SPOTLIGHT spends 56.00 seconds in failover, compared to a median downtime of 60.76 seconds for Totoro+ and 136.94 seconds for FedGuard.

Totoro+’s recovery mechanism relies on two things: state replication and child-triggered rejoin. Each round, the master replicates its state to its $k=2$ nearest neighborhood nodes; on failure, the child detects it, routes a JOIN using the AppId, and the closest node in NodeId space becomes the new master and pulls the replicated state. The authors test this mechanism

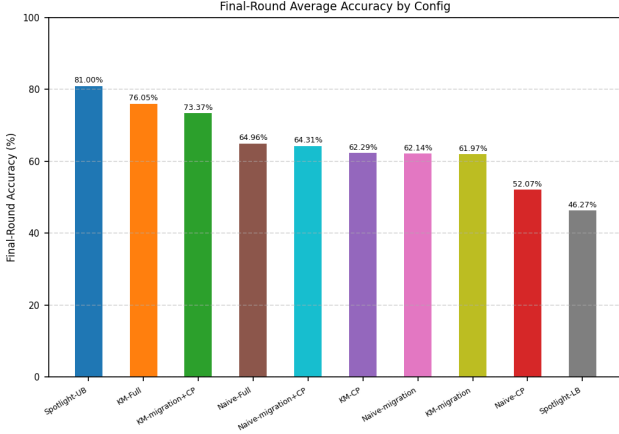


Figure 31: Average accuracy across fault tolerance configurations under non-IID client data (10 clients, Dirichlet $\alpha = 1.5$).

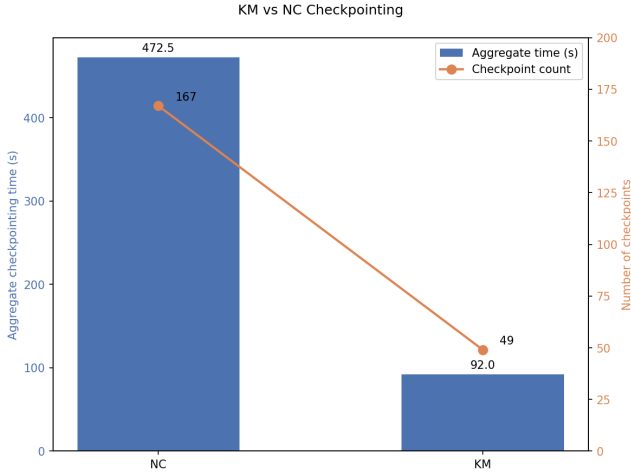


Figure 32: Aggregate checkpoint time and checkpoint count: KM-CP vs. Naive-CP.

at 5% simultaneous node failure; under our trace, preemptions are far more frequent, and progress is interrupted before each recovered master accumulates meaningful training, so accuracy stalls even though state is never lost.

FedGuard persists model state via ZooKeeper-backed checkpointing and its super run abstraction, so weights are not lost on leader failure. However, each leadership change still triggers a de-structuring and reinitialization of the Flower federation itself, since a new leader means a new Flower run. This reinitialization overhead, not weight loss, is what drives up time between preemptions before FedGuard resumes meaningful progress. These limitations do not exist in SPOTLIGHT, allowing it to significantly outperform the other baselines in recovery as well accuracy.

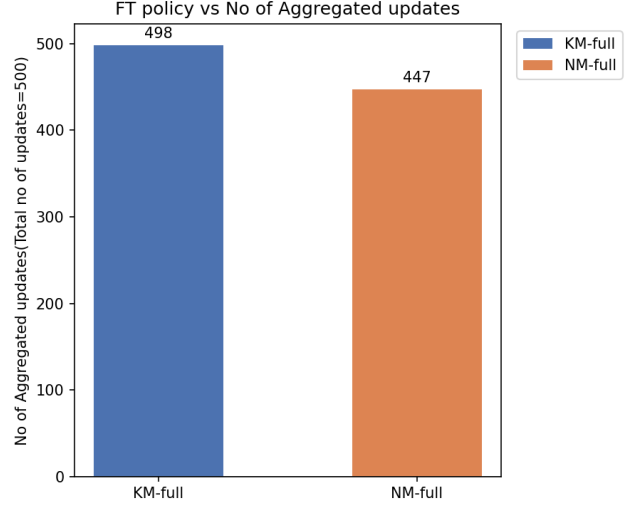


Figure 33: Number of updates received by SPOTLIGHT under the KM-full and the NM-full policy, out of 500 total updates sent by real clients.

Config	Compression	Failures/Hr
Base	10x	13
Medium	30x	41
High	60x	83

Table 9: Spot Emulator compression ratio and expected failure frequency for each configuration.

7 Discussion

We now discuss three limitations of our approach: the assumption of uniform failure risk across aggregators, the cost of full-model checkpointing under preemption, and the structural overhead of a static hierarchy.

Uniform failure risk across zones. SPOTLIGHT assumes that all aggregators are deployed within a single zone, under which migrating an aggregator to a healthy replica incurs low overhead. This assumption breaks down once aggregators are spread across multiple zones. The migrator currently treats any available aggregator as an equally valid migration target, but preemption rates differ substantially by zone, as shown in Figure 46. If the target zone is itself high-risk, migrating to it is a poor tradeoff: the aggregator may be preempted again shortly after, and each migration already carries non-trivial cost from cross-zone bandwidth limits and the latency of re-synchronizing FL state. A zone-aware migrator would need to weigh a candidate target’s preemption risk against this migration cost, rather than swapping to the first available aggregator regardless of its zone.

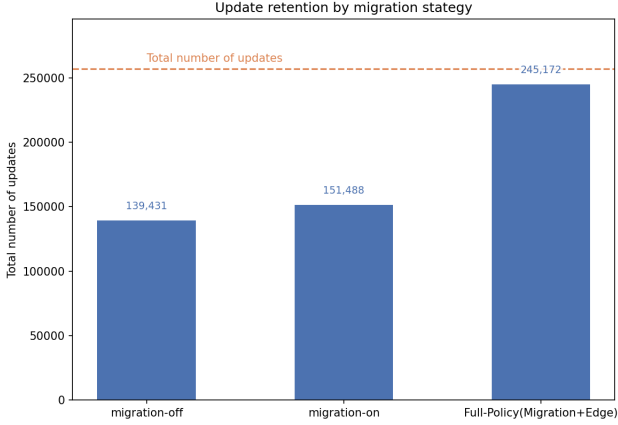


Figure 34: Total number of updates describes the number of updates from that were successfully received and aggregated by SPOTLIGHT under different FT configurations

Full-model checkpointing under preemption. SPOTLIGHT checkpoints the full model at each checkpoint trigger, rather than checkpointing only the parameters that changed since the last checkpoint, an approach shown to reduce checkpoint write bandwidth for large models such as production recommendation models that can reach terabytes in size [16]. This approach works by storing a differential checkpoint relative to a baseline checkpoint, reducing what needs to be written each interval. However, it depends on that baseline being durably persisted; reconstructing the model requires reading both the baseline and the latest delta. On a spot instance, the baseline itself may not finish writing before the instance is preempted, in which case every delta computed against it becomes unusable. This makes differential checkpointing, despite being effective for large models on stable clusters, unsuitable for our setting without further changes.

Structural cost of a static hierarchy. SPOTLIGHT maintains a hierarchical aggregation structure regardless of the number of active aggregators, even though the hierarchy is only necessary once multiple aggregators must be coordinated. Figure 15 shows replica count over time under the KTH trace with FedAvg: the system operates with a single replica, for which a flat structure would suffice, for 48.3% of the observed trace, and scales to two or more replicas for the remaining 51.7% of the time. Once the system reaches a single-replica state, it could in principle collapse to a flat structure and improve cost efficiency, but this gain is offset by two costs: VM startup latency when scaling back into a hierarchy, and the synchronization overhead of migrating aggregator state across the structural change. Realizing this tradeoff in practice would require an optimization criterion that weighs the savings from operating flat against these switching costs before deciding to change structure.

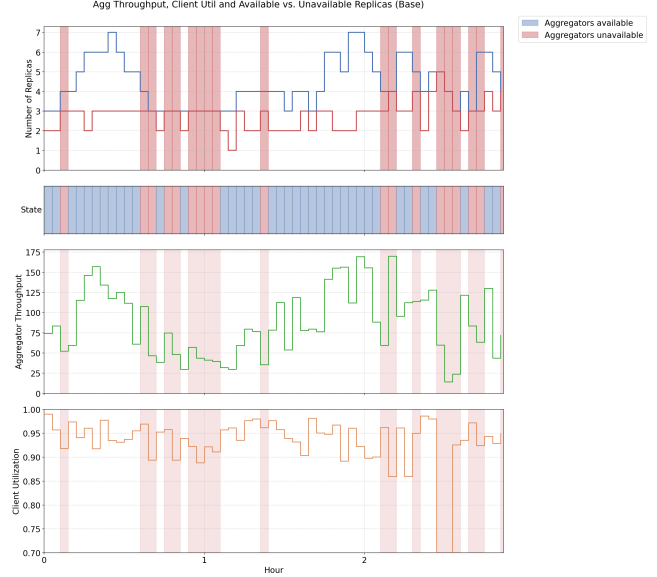


Figure 35: Replica availability, aggregator throughput, and client utilization over time under the base configuration on the KTH trace. In the top panel, the blue line tracks the number of available replicas, rising and falling as the Scaler adds and removes replicas in response to client demand, and the red line tracks the number of unavailable replicas; shaded regions mark periods where the two lines meet, indicating cloud aggregators are unavailable and SPOTLIGHT falls back to failover aggregators.

8 Related Work

This section discusses existing FL systems and their limitations with respect to scalability, fault tolerance and cost efficiency.

FL systems must support a large number of participating clients. Existing systems such as Oort, Flower, and Papaya, among others [2, 6, 8–10, 17, 24, 26, 28], are designed as single-aggregator systems where the aggregator is responsible for all functions in the system, such as aggregating updates, client coordination, and distributing the updated model. Single-aggregator systems are bottlenecked in their ability to support large numbers of clients due to their compute and network resource limits, while client selection techniques are used in systems such as Oort and PyramidFL [26, 28] that select the number of clients whose updates the aggregator incorporates into the global model. However, even with client selection, the resource limitations of single-aggregator systems will continue to be a bottleneck. In contrast to single-aggregator systems, hierarchical systems distribute aggregation across multiple tiers, usually by introducing a tier close to the client before sending updates from the client to the cloud aggregator. Existing hierarchical systems [1, 14, 31, 34, 35, 43, 54, 55, 60] are statically provisioned at deployment. This static provision-

Base: Availability vs. Unavailability (% of Total Time)

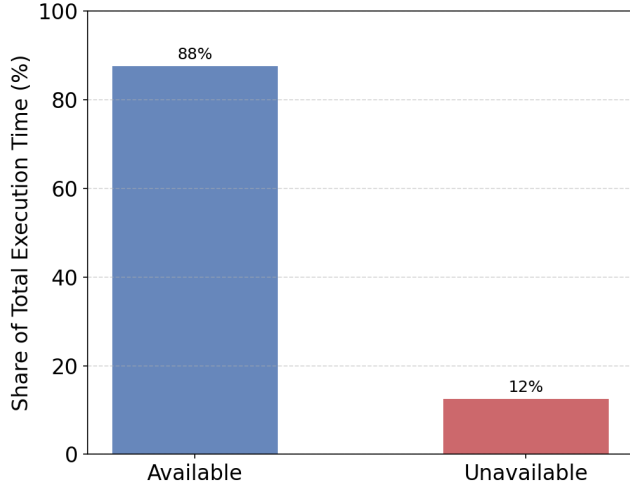


Figure 36: Share of total execution time SPOTLIGHT spent with cloud aggregators available versus unavailable, under the base configuration.

ing causes the systems to eventually form the same bottleneck faced by single-aggregator systems. Scaling the aggregator, as discussed in the previous section, is a non-trivial challenge. P2P-FL systems such as D2, ByRDIE, BRIDGE, AsyDFL, BrainTorrent, and Totoro+ [11, 30, 45, 50, 58, 59] overcome the bottlenecks faced by single- and hierarchical-aggregator systems by having the client and aggregator reside on the same device within the network. In P2P-FL systems, the client and aggregator roles are played by the same device: devices share updates with their neighbors while aggregating. Totoro+ [11] deviates from this design by forming a locality-assigned tree, where the aggregator waits for its assigned workers to send their updates. While P2P systems are not affected by aggregator bottlenecks, device heterogeneity and the coordination overhead of maintaining model consistency affect their ability to scale.

FL systems are typically deployed on on-demand instances in order to provide an aggregator that is always available. Serverless FL systems deviate from this model by provisioning the aggregator only when required to aggregate, allowing them to scale elastically to meet client demand [19, 20, 42]. The aggregators are stateless, so it must pull and store the global model from external storage on every invocation. This limitation compounds when using stateful algorithms such as FedAdam, where the optimizer state must also be stored and retrieved externally, imposing a coordination overhead that is not present in on-demand systems.

Spot instances offer a cost-efficient alternative to on-demand instances but can be reclaimed by the cloud provider at any time, priced at a significant discount to reflect that risk.

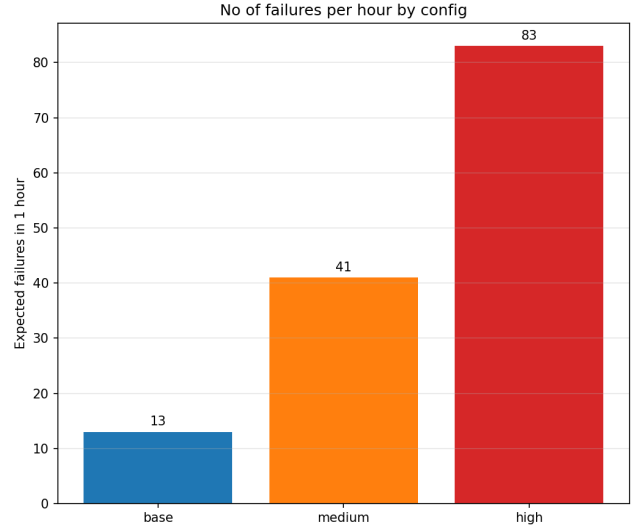


Figure 37: Expected failures per hour by configuration, as compression ratio increases.

Existing systems [4, 15, 16, 37, 46, 51, 61] utilize spot instances for inference and training of DL workloads. In the case of FL systems, the aggregator plays a central role in the functioning of the system: the aggregator forms a single point of failure (SPOF), with its failure causing complete failure of the FL system. Existing FL systems such as Raft-FL and FedGuard [5, 12] address this limitation with a consensus mechanism that elects a new aggregator to act as the leader while the remaining aggregators act as standby, but this failure model is built for the reliable, on-demand infrastructure these systems assume, where failures are rare and affect only one aggregator at a time. It is not suitable for spot instances, where preemptions can affect multiple aggregators concurrently, leading to stalled elections and increased downtime. In addition to the limitation of consensus mechanisms, these systems are also reactive in nature and trigger migration only after aggregator failure is detected. In the case of FL systems, this also contributes to downtime and often requires a restart of FL jobs on the new aggregator.

Among the existing systems surveyed, none are scalable, cost-efficient, and fault-tolerant without incurring downtime. SPOTLIGHT addresses this gap as a scalable, hierarchical FL system that can be deployed on spot instances with fault tolerance that ensures aggregator failures do not degrade FL system performance. Table 11 summarizes these limitations across representative systems from each architecture discussed above.

9 Conclusion

We present SPOTLIGHT, a federated learning system that scales elastically in response to client demand on Spot in-

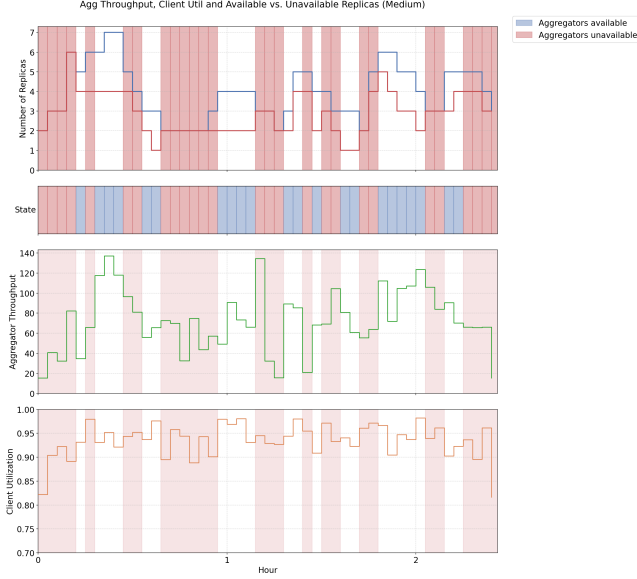


Figure 38: Replica availability, aggregator throughput, and client utilization over time under the medium configuration for KTH trace

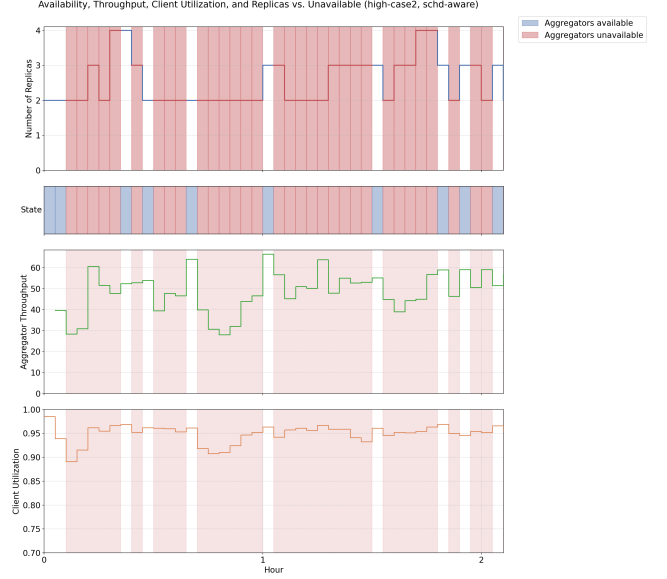


Figure 39: Replica availability, aggregator throughput, and client utilization over time under the high configuration for KTH trace

stances while managing Spot preemptions through its fault tolerance policies. §6.1 shows SPOTLIGHT sustains high client utilization, a median of 0.98 across four aggregation algorithms (Table 6), by scaling in response to load and using its scheduler to exploit added aggregator capacity, at lower cost than Papaya on the KTH trace, 49.5% under FedAvg and 21.8% under FedAdam (§6.1.2).

§6.2 shows SPOTLIGHT’s KM-Full policy outperforms every other fault tolerance configuration tested, reaching 81.52% accuracy under IID and 76.79% under non-IID client data, ahead of the next-best configuration by 9.7 and 3.4 percentage points (Figures 30, 31). It does this while checkpointing the least of any configuration tested (Figure 32), proactively migrating aggregator state ahead of predicted failure, and falling back to failover aggregators only when no cloud aggregator is available.

§6.3 evaluates this policy under increasing failure frequency, and the key result is that its cost is disproportionate: client utilization degrades far less than the failure rate would suggest. As unavailable time rises from 12% at base to 67% at high (Figure 40), aggregator throughput takes the larger hit, since failover aggregators cannot match cloud aggregator capacity, while client utilization stays comparatively close to its fault-free level throughout (Figure 41). This asymmetry is also visible in recovery: 77% of outages recover to at least 70% of pre-outage throughput at the base configuration, 74% at medium, and 38% at high (Figure 42), meaning throughput recovery weakens well before client-facing progress does.

Taken together, SPOTLIGHT absorbs a large increase in failure frequency at a comparatively small cost to the metric

that determines whether the FL process keeps moving: client-side utilization degrades far less than aggregator throughput and recovery time do, showing the system’s fault tolerance policy protects the right thing as failures increase..

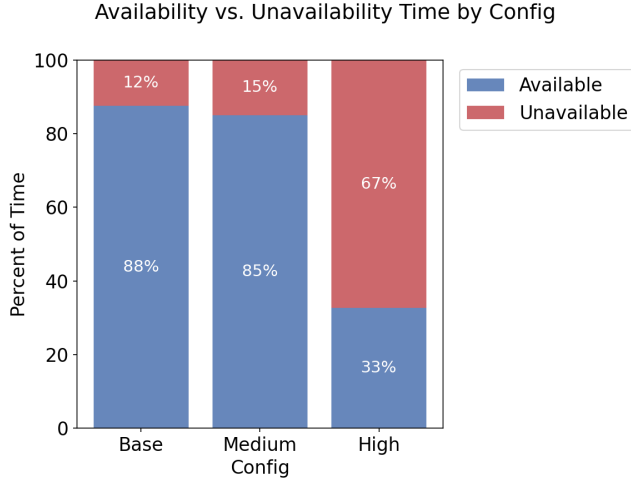


Figure 40: Share of total execution time SPOTLIGHT spent with cloud aggregators available versus unavailable, by configuration.

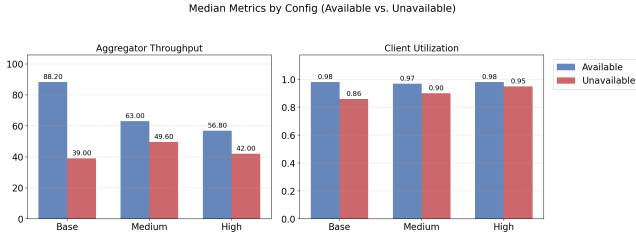


Figure 41: Median aggregator throughput and client utilization, available versus unavailable, by configuration.

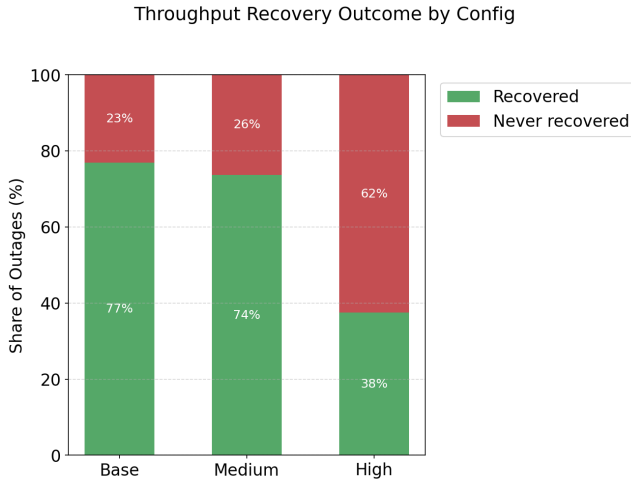


Figure 42: Share of outages classified as recovered or never recovered, by configuration.

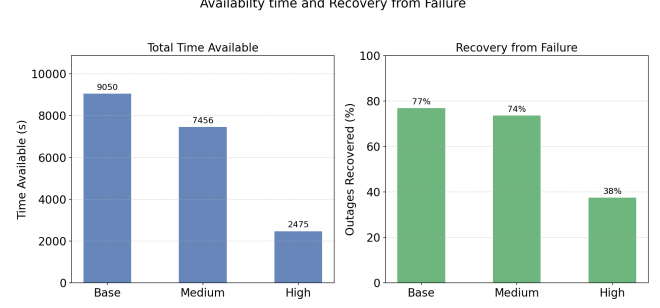


Figure 43: Total time available and outage recovery rate, by configuration.

Table 10: Fault tolerance ablation configurations

Configuration	Description
Spotlight-UB	No faults injected (upper bound)
Spotlight-LB	Faults injected, no recovery (lower bound)
Naive-CP	Checkpoint on availability only
Naive-migration	Migrate only after aggregator failure
Naive-migration+CP	Naive migration combined with naive checkpointing, no failover offloading
Naive-Full	Naive checkpointing + naive migration + failover aggregator offloading
KM-CP	KM checkpointing only
KM-migration	KM migration only
KM-migration+CP	KM migration and checkpointing, no failover offloading
KM-Full	KM checkpointing + KM migration + failover aggregator offloading, with the KM configuration governing all FT decisions

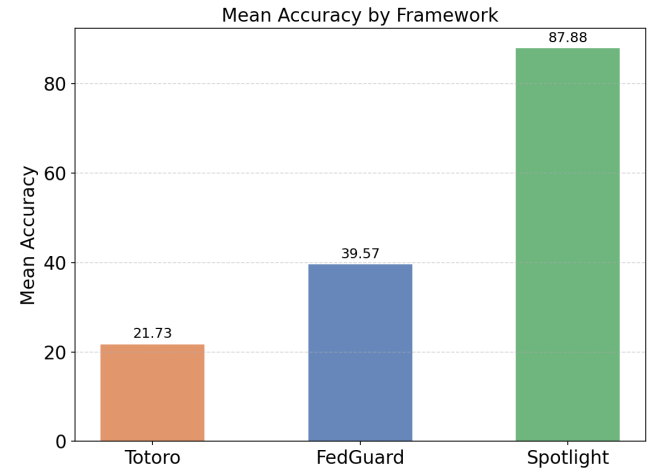


Figure 44: Mean accuracy under preemption for SPOTLIGHT, Totoro+, and FedGuard on IID CIFAR-10.

Table 11: Comparison of representative FL systems against SPOTLIGHT.

System	Architecture	Platform	Instance Type	Scalability	Spot-Compatible	Fault Tolerance
Oort [26]	Single-aggregator	Data-center	On-demand	×	×	×
Flower [6]	Single-aggregator	Data-center	On-demand	×	×	×
Papaya [17]	Single-aggregator	Data-center	On-demand	×	×	×
HiFlash [55]	Hierarchical	Data-center	On-demand	×	×	×
Totoro+ [11]	P2P	Edge	N/A	~	×	×
AdaFed [20]	Serverless	Data-center	On-demand	✓	×	×
Raft-FL [12]	Single-aggregator (consensus)	Data-center	On-demand	×	×	×
FedGuard [5]	Single-aggregator (consensus)	Data-center	On-demand	×	×	×
SPOTLIGHT	Hierarchical	Data-center	Spot	✓	✓	✓

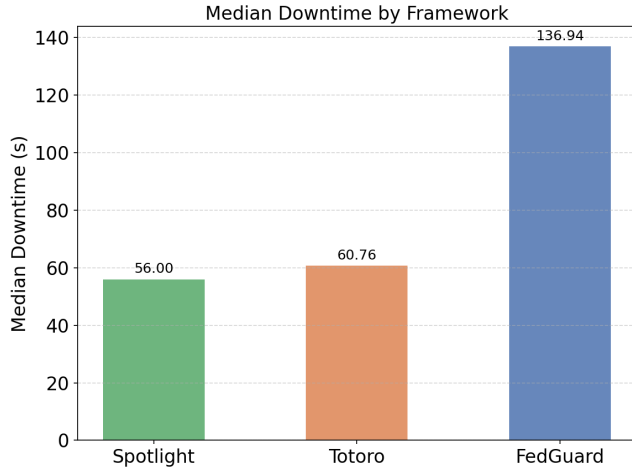


Figure 45: Median downtime (SPOTLIGHT: average time in failover) for SPOTLIGHT, Totoro+, and FedGuard.

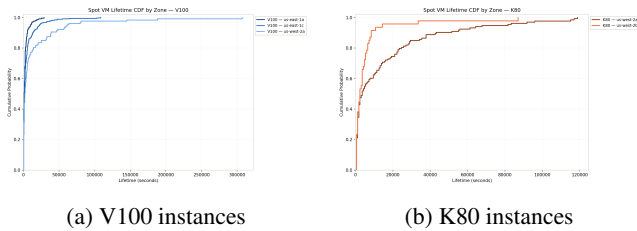


Figure 46: Kaplan-Meier estimates of spot VM lifetime by availability zone, computed from traces released with [56].

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